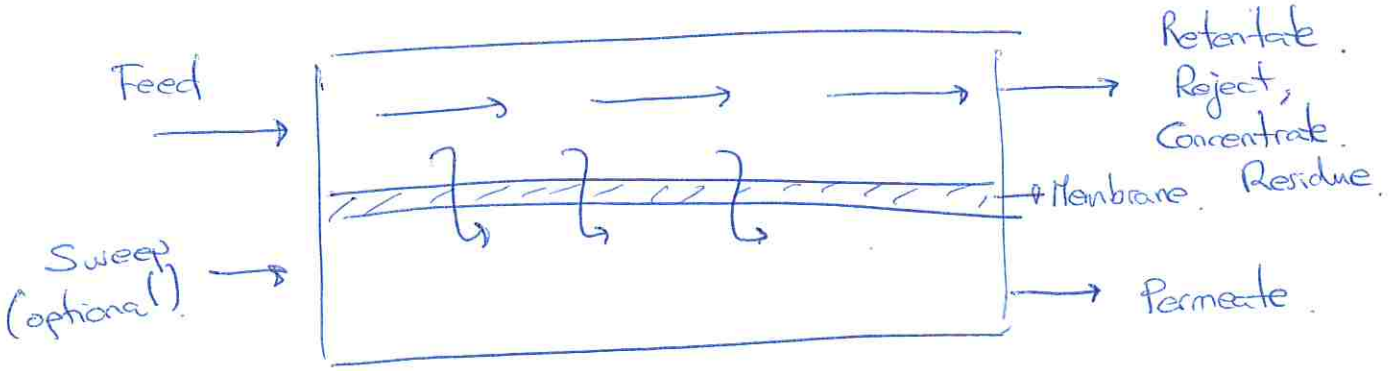
4. MEMBRANE SEPARATIONS

Membrane separation processes are not operated as equilibrium-staged processes. Instead, these separations are based on the rate at which solutes transfer through a membrane.

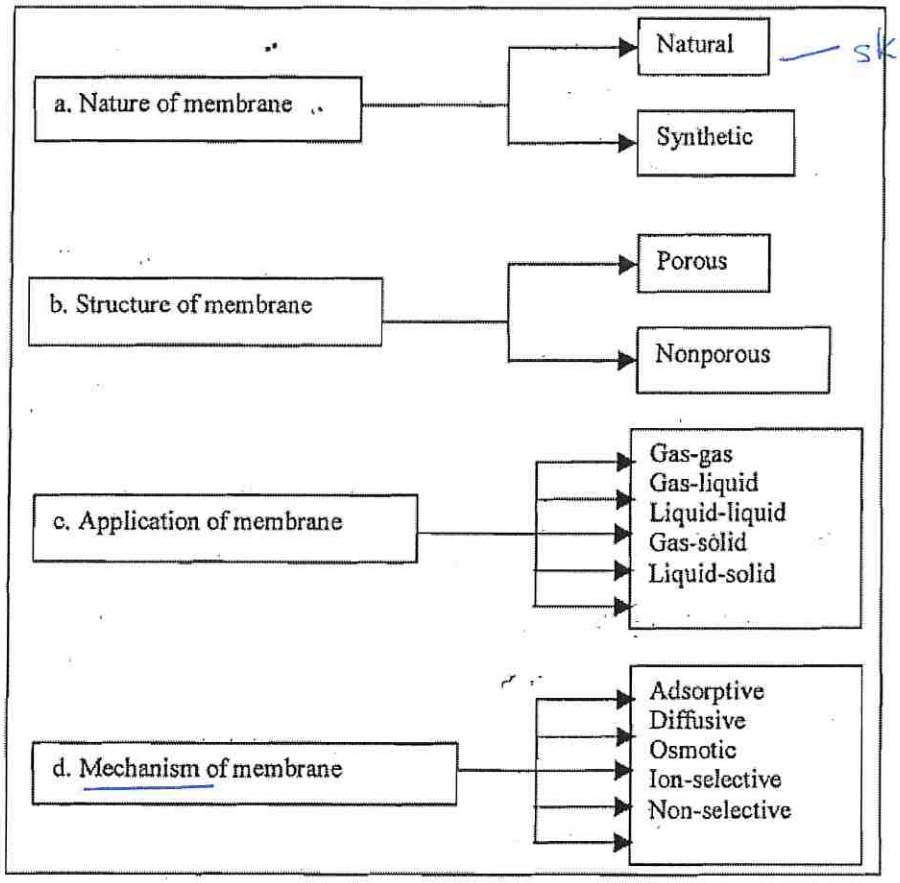
The separation of *gaseous* or *liquid* mixtures using *semi-permeable membranes* allows some constituents of the mixture to pass through the membrane more readily than others.

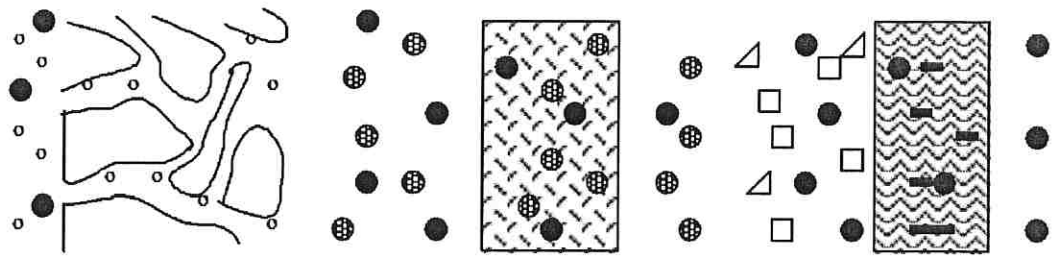


- The membrane represents a *selective* barrier between two phases and its structure determines the nature of the selectivity in the separation process.
- Membrane *structure* is controlled by its manufacturing process.
- The membranes may be thin layers of a rigid material such as porous glass or sintered metal, but more often are flexible films of synthetic polymers with a high permeability for certain types of molecules.

4.1 Types of membrane systems

The membranes can be classified according to different parameters:





porous membrane [micro/ultrafiltration]

nonporous membrane [gas separation pervaporation/

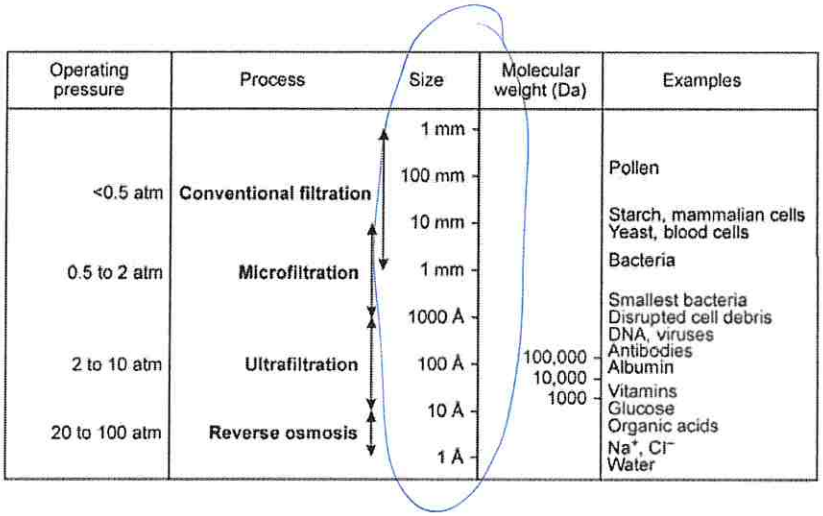
Liquid membrane [carrier mediated transport]

Transport of species through the membrane involves driving forces such as pressure gradients or concentration gradients.

Membrane processes are called a variety of names, depending on the nature of the phases on either side of the membrane and the size of the species passing through it.

Membrane Process	Phase 1	Phase 2	Driving Force	Application	Membrane Nature
Microfiltration	L	L	ΔP (0.1–2 bar)	Solids removal or concentration	Porous inorganics/ organics
Ultrafiltration	L	L	ΔP (1–5 bar)	Macromolecular concentration	Polymerics
Hyperfiltration (Reverse Osmosis)	L	L	ΔP (10–100 bar)	Desalination	Asymmetric or composite polymers
Gas Separation	G	G	ΔP (1-20 bar)	Hydrogen recovery	Dense elastomers
Dialysis	L	L	ΔC	Artificial kidneys	Hydrophilic polymers
Pervaporation	L	G	ΔP	Solvent dehydration	Elastomeric and glassy polymers
Electrodialysis	L	L	ΔE	Ion separation	Non porous polymers
Membrane Distillation	L	L	ΔT or ΔP	Pure water	a/symmetric porous

They are also used across many length-scales:



4.2 Advantages and disadvantages of membrane separations

Replacement of more-common separation operations with membrane separations has the potential to save energy and lower costs. It requires the production of high mass-transfer flux, defect-free, long-life membranes on a large scale and the fabrication of the membrane into compact, economical modules of high surface area per unit volume.

It also requires considerable clean-up process feeds and careful control of operating conditions to prevent membrane deterioration and to avoid degradation of membrane functionality due to caking, plugging and fouling.

Advantages	Drawbacks
Batch or Continuous operation	Concentration polarisation
Low energy consumption	Membrane fouling
Easily combined with other separation processes	Low membrane lifetime
Mild separation conditions	Limited selectivity
No additives required	Need pilot-scale experiments
Membrane properties can be tailored to situation	

4.3 Membrane structure and separator arrangements

The flux through a membrane is inversely proportional to its thickness, hence ideal membrane is:

- thin
- does not contain *any* holes or weak-spots
- structurally strong to withstand pressure drops

To achieve this, many membranes are asymmetric or composite in nature with a thin membrane (0.1–1 μm thick) being supported by a highly porous thicker layer (50–200 μm thick) capable of withstanding the pressure drops but offering little resistance to flow.

Supports are normally porous ceramic metals or polymers, and special methods of preparation are needed to formulate a thin dense layer (or skin) on one side (which confers the selectivity)

Asymmetric membranes can be prepared in the form of flat sheets, tubes, or hollow fibres with internal diameters as small as 40 μm. Flat sheet membranes for reverse osmosis are usually used in spiral-wound membrane modules to increase surface area per unit volume.

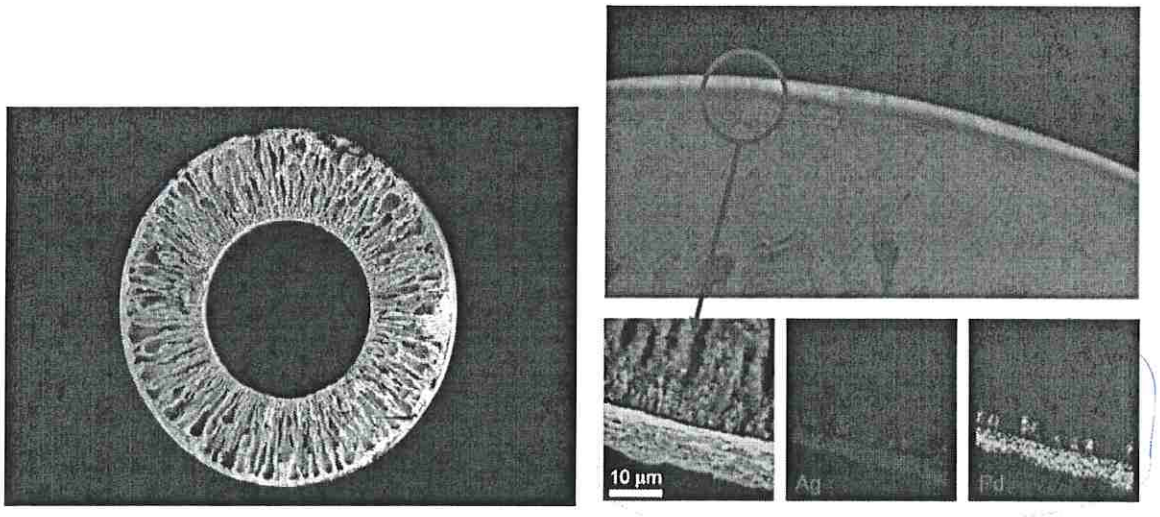


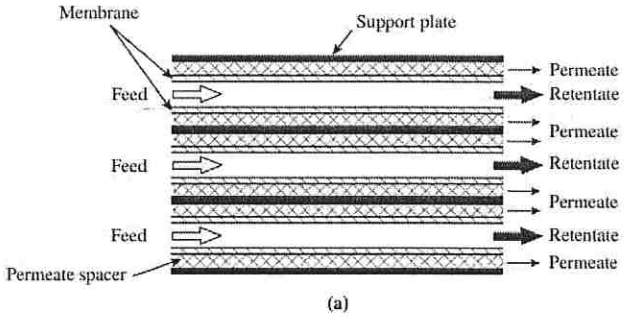
Figure: Cross section of hollow fibre support (left) for a continuous Ag-Pd metallic membrane for selective hydrogen separation

4.4 MEMBRANE FILTRATION EQUIPMENT

Membrane filtration is normally carried out with **modular equipment**. Modules allow large surface area for filtration per unit volume. To achieve large fluid velocities and contact area with the membrane, the feed is normally divided into many narrow channels/layers encased by membrane.

Various configurations exist:

- **Flat Sheets**: Membrane is housed in plate-and-frame or cassette devices. The separation between the membranes is ~0.5 and 2.0 mm. It can achieve membrane surface areas per unit volume of ~ 300-500 m^2/m^3.



- **Spiral**: consist of pairs of flat membranes wound in a spiral separated and supported by mesh screens, rolled and packed in a tube.
  The feed moves along the length of the tube and retentate comes out on the other side. The permeate moves towards the center of the tube following the spiral pattern and comes out through dedicated outlets. The membrane surface area per unit volume is ~ ~ 600-1000 m^2/m^3.

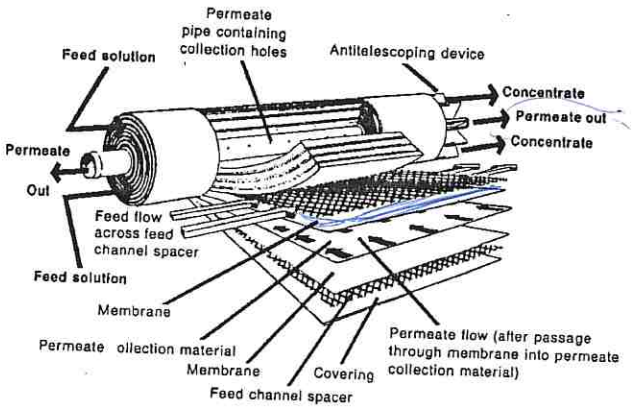
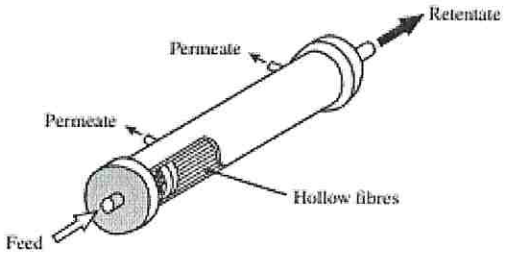


Figure: A spiral wound membrane module

- **Hollow Fibers:** the modules typically contain 50-5000 membrane tubes, with diameters between 0.2 and 2 mm. It presents the highest membrane surface area per unit volume (~ 1,200m^2/m^3), however, it is susceptible to blockage due to narrow tubes.

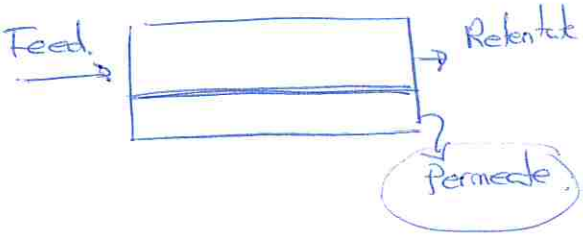


A commercial hollow fibre membrane separator might contain ~ 1 million hollow fibres aligned in a parallel manner in a shell of up to 10–30 cm diameter and up to 3–5 m in length, in order to achieve a high interfacial area per unit volume.

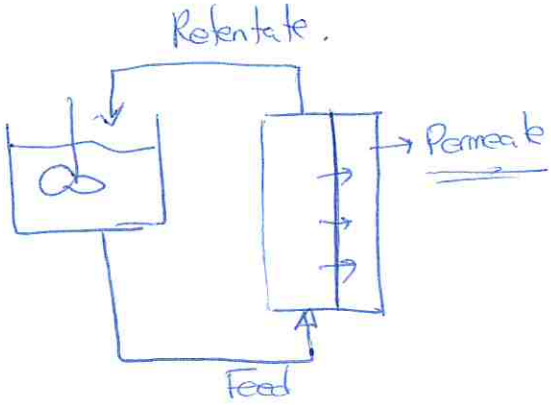
4.5 MEMBRANES MODES OF OPERATION

Membrane filtration can be carried out using various process configurations and modes of operation:

i. **Single pass filtration:** It does not involve the recirculation of the retentate. Because in membrane filtration permeate flux is usually very low compared with cross-flow rates (e.g. feed), only small values of volume concentration ratio (VCR = $V_{in}/V_{permeate}$) can be obtained. This configuration is of limited utility.

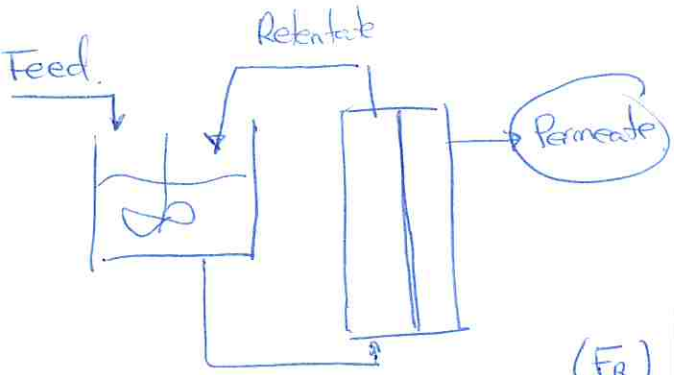


ii. **Batch filtration:** in this case, the retentate is re-circulated. It is often used in applications where the permeate contains the target product. It is unsuitable for concentration of retained products that might be damaged by repeated pumping (in particular biological samples). The volume of the retentate decreases with time, allowing to achieve large VCR.



Because the concentration of solute in the feed increases with time, so does the resistance to flow. Thus, the process becomes progressively less efficient.

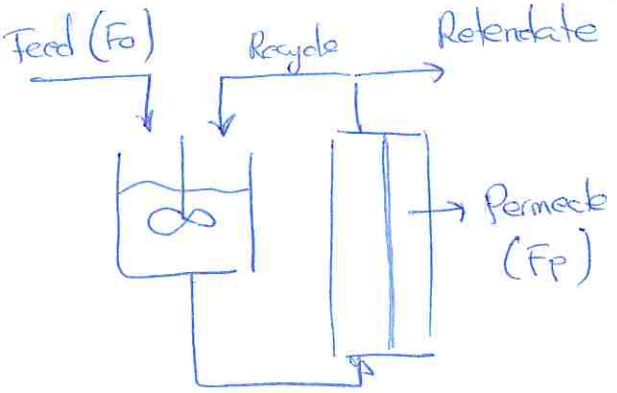
iii. **Fed batch:** Fresh feed is continuously added during the operation of a batch filtration, extending the number of times retentate is recirculated, increasing the risk of damage of biological samples. It is commonly used in industry when the desired product is in the permeate.



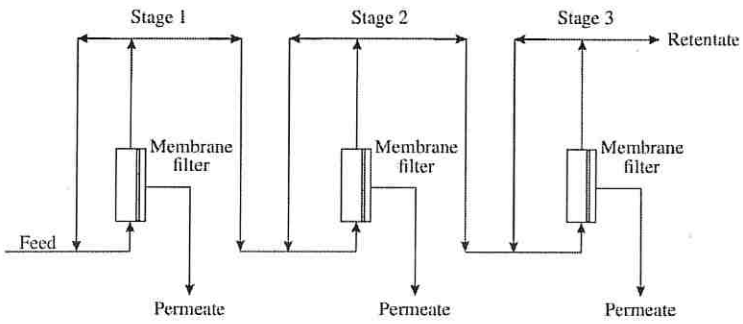
iv. **Continuous:** Fresh feed is added at volumetric flow rate (F_0) while retentate is removed at volumetric flow rate F_R. The volume concentration ratio is then given by:

$$VCR = \frac{F_0}{F_R} = \frac{F_R + F_P}{F_R}$$

This is the most common approach in industry for large-scale membrane filtration.



v. **Multistage:** It consists of a cascade of continuous filtration operations. Units are operated in series with respect to retentate flow, and in parallel with respect to permeate flow. This configuration is often used when membranes with different characteristics (e.g. cutoff sizes) are needed. For instance, microfiltration followed by ultrafiltration.



Almost all commercial operations are performed in a (semi)-continuous mode, with crossflow over the surface of the membrane to increase fluxes, rather than in a dead-end, batch mode.

A large module may contain several thousand m^2 of membrane area, but for many applications, multiple units are required, usually arranged in parallel.

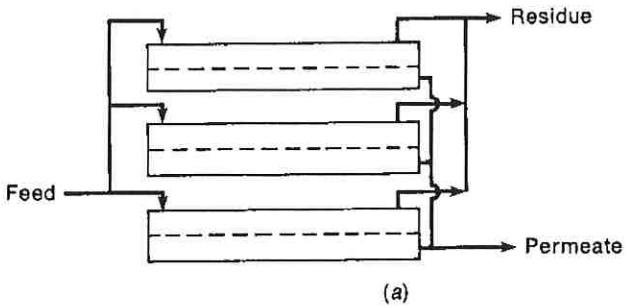
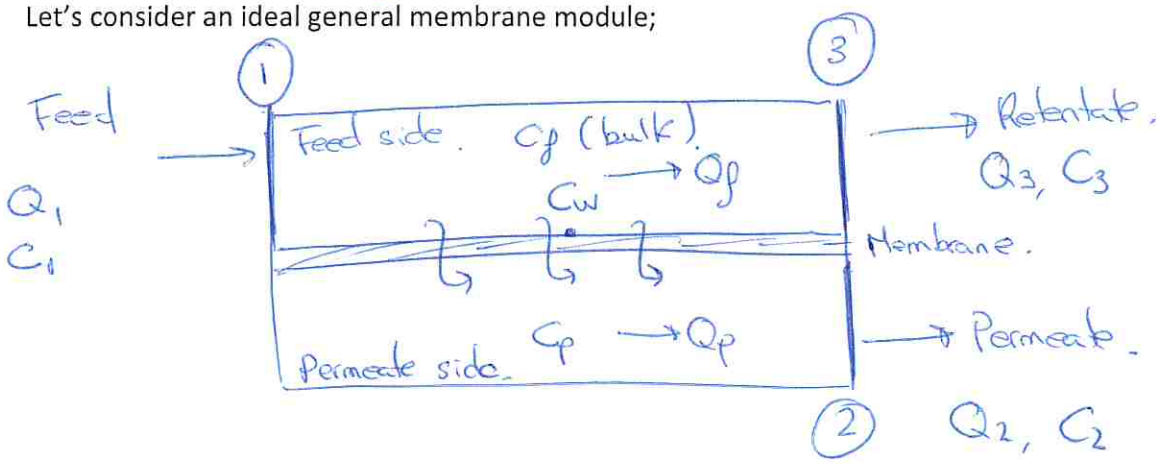


Figure: Parallel arrangement of membrane modules

4.6 NOMENCLATURE AND DEFINITIONS

Let's consider an ideal general membrane module;



Where:

- Q is a volumetric flow rate,
- C is a concentration of solute: C_w is concentration of solute at the surface of the membrane on its feed side.
- All the parameters Q_f, C_f, Q_p, C_p may change along the length of the membrane.

Flux

The flux of permeate, *J*, through a membrane is expressed as:

$$\left(\frac{m^3}{m^2 \cdot s}\right) \quad J = \frac{\text{Transfer rate}}{\text{Transfer area}} = \frac{Q_p}{A}$$

Where: Q_p is the volumetric flow rate of permeate (m^3/s)

A is the area of the membrane

One can also define a molar flux of permeate, *Np* [$kmol/m^2s$]

$$\left(\frac{kmol}{m^2 \cdot s}\right) \quad N_p = \frac{Q_p \cdot \rho_m}{A}$$

Where: ρ_m is the molar density of the permeate flow ($kmol/m^3$).

In liquid systems the total molar concentration C_T is often used instead of ρ_m.

Selectivity and retention

The passage of a component is determined by the interaction between the component and the membrane. This gives rise to *selectivity* in the separation of mixtures of components.

In gas separations, the selectivity is normally expressed as a ratio of the permeabilities of the membrane to the two components, K_{mA} and K_{mB}. This is an analogous situation to the concept of relative volatility.

$$\alpha = \frac{K_{m,A}}{K_{m,B}} \quad \text{(permeability)} \qquad (4.1)$$

In ultrafiltration (UF) and reversed osmosis (RO), selectivity is expressed by the 'retention' of a solute or its 'rejection' by the membrane, expressed as a solute retention fraction, *R*.

$$\text{Retention} \equiv R = \frac{C_f - C_p}{C_f} \qquad (4.2)$$

Apparent retention fraction: based on bulk conditions on the feed side of the membrane (i.e. C_f) (as expressed in Eqn. 4.2).

True retention fraction: based on conditions at the actual membrane surface (*i.e. Cw* is used in Eqn. 4.2 in place of *Cf*).

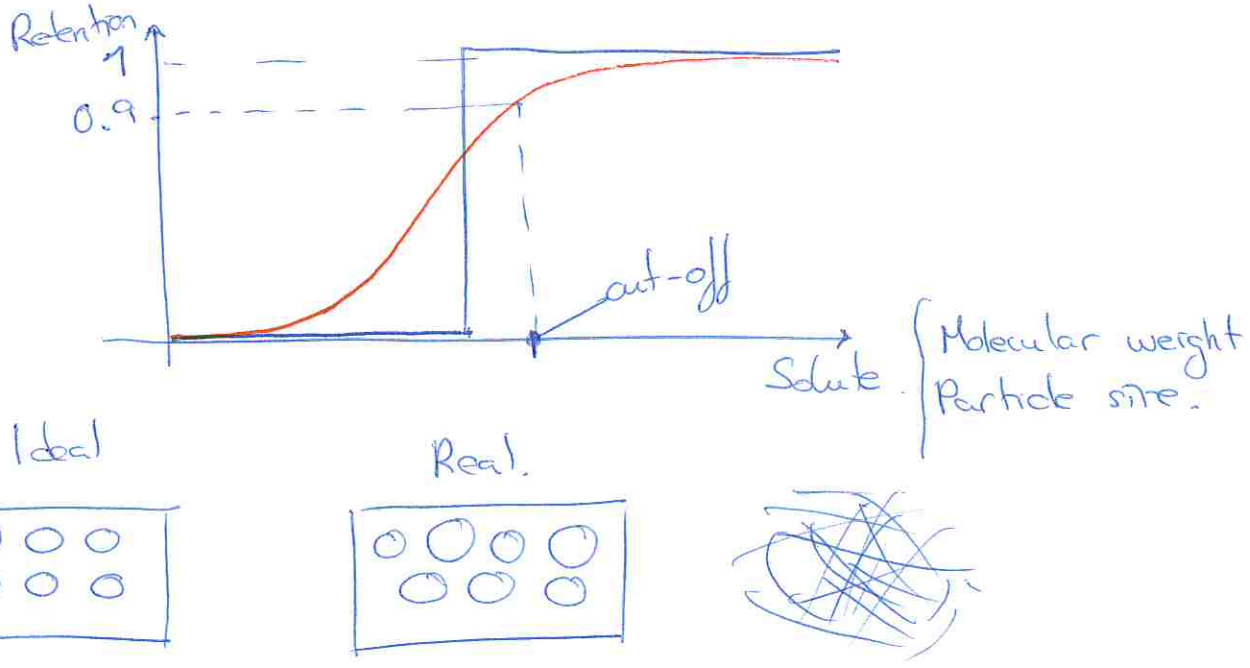
$$R_{true} = \frac{C_w - C_p}{C_w}$$

The value of the retention *R* can be affected significantly by *concentration polarisation* and other flux effects (see later).

A membrane which rejects a species completely ($C_p = 0$) has R = 1.

Porous membranes are often characterised by a *cut off molecular weight*:

- the relative molecular mass of solutes at which 90% rejection by the membrane occurs.
- This value contains no information about the nature of the cut off or the shape of the membrane "holes" or the shape of the excluded species; membranes are tested on the lab scale to determine their performance in new applications.



The recovery, *S*, is defined as the ratio of permeate and inlet feed flow rates (volumetric);

$$S = \frac{Q_p}{Q_1} \rightarrow \text{feed flowrate.} \qquad (4.3)$$

Thus, *S* varies along the length of the membrane. The *overall recovery* achieved on reaching the outlet is

$$S_2 = \frac{Q_2}{Q_1}$$

4.7 TRANSPORT EQUATIONS

The rate of transport of a component through a membrane will be determined by:

- a driving force, *and*
- a resistance to mass transfer

Formally, the driving force is the difference in the fugacities of a component between each side of the membrane, commonly approximated as the difference in:

- Concentrations, ΔC_j, for liquids
- Partial pressures, Δp_j for gases

The mechanism of transport through the membrane determines the nature of the transfer resistance.

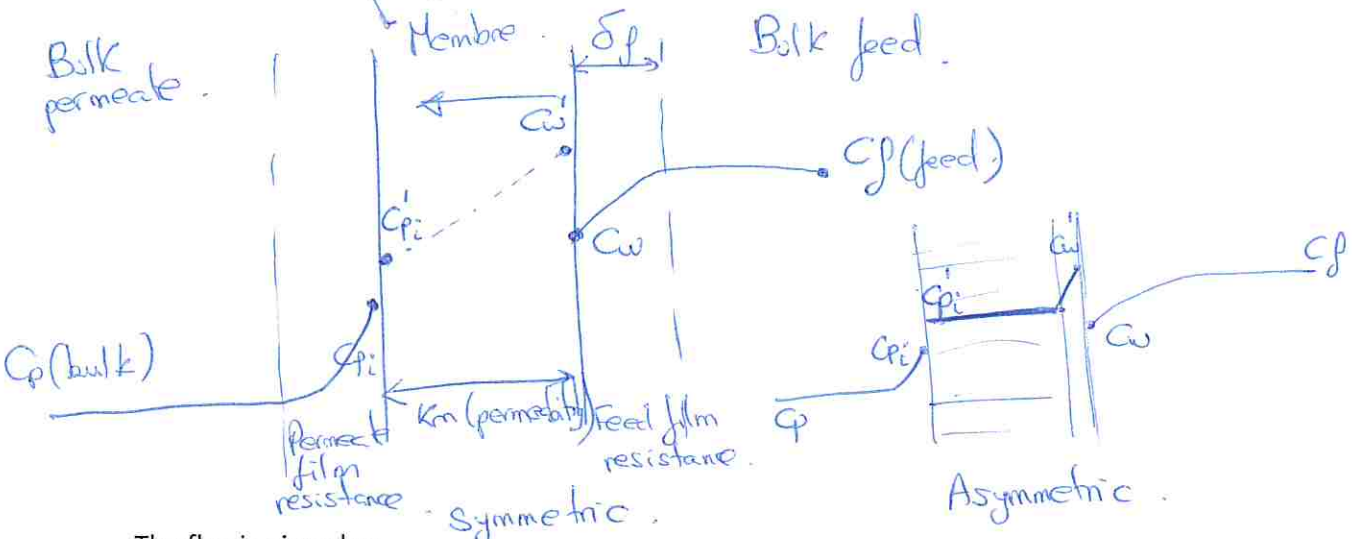
In liquid separations there is bulk flow of solvent through the membrane driven by the difference in the pressures, ΔP, on either side of the membrane. $\Delta P = P_f - P_p$

Resistances to mass transfer

The transport of component A through a membrane can be modelled as a series of mass transfer resistances. In the absence of any concentration polarisation ($C_w < C_f$)

For symmetric membranes, the resistance of the membrane is uniform (isotropic).

For asymmetric membranes, the resistance of the membrane is concentrated in a thin, "active" layer that is supported by an easily penetrated (but non-selective) porous matrix.



The flux is given by:

$$N_j = K_{O,j}\left(C_{f,j} - C_{p,j}\right) \tag{4.4}$$

where $K_{O,j}$ is the overall permeability of the system for component *j*

$$\frac{1}{K_{O,j}} = \frac{1}{k_{f,j}} + \frac{1}{K_{m,j}} + \frac{1}{k_{p,j}} \tag{4.5}$$

Note the addition of the reciprocals of the individual contributions to mass transfer resistance (analogous to heat transfer resistances in series).

- In gas separation the membrane coefficient, K_m, dominates the situation, so:

$$K_{O,j} = K_{m,j}$$

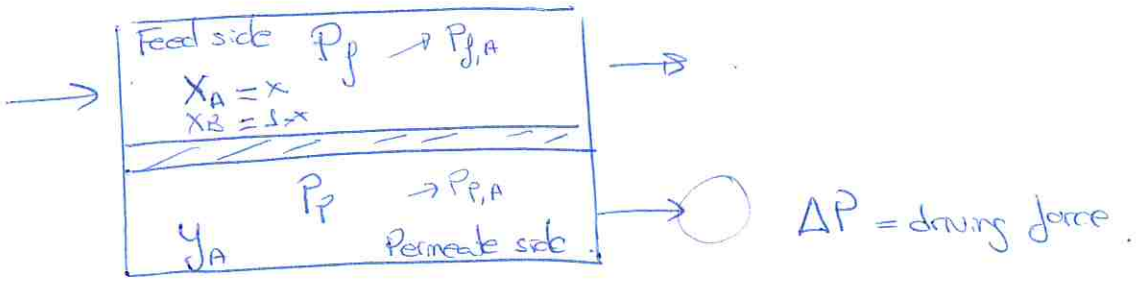
- In liquid phase separations where *concentration polarisation* occurs, the feed-side mass transfer resistance becomes the major resistance to mass transfer.

4.8 MEMBRANE SEPARATIONS IN GAS SYSTEMS

Gas permeation membrane systems are used commercially for separation of permanent gases. Some common applications are purification of helium, hydrogen and carbon dioxide. They are also used for the production of high purity nitrogen from air and production of air enriched in oxygen.

Let's consider a binary gas mixture with components A and B. Let's also consider the following assumptions:

- The membrane is chosen such that $K_{m,A} \neq K_{m,B}$ and these permeabilities will be assumed to be not functions of composition or concentration (this may not be true in practice). It is highly likely that both A and B can pass through the membrane.
- There are no film mass transfer resistance effects i.e. $K_{O,j} = K_{m,j}$ for all j



Molar flux of each component through the membrane can be expressed using partial pressure differences as driving forces.

Note: typical, appropriate units for K_{mj} in these formulations are kmol/(m^2 s bar)

$\sim £200/m^2$ Binary mixture $X_A = x$ $X_B = 1-x$

$$N_A = K_{m,A}\left(P_{f,A} - P_{p,A}\right) = K_{m,A}\left(P_f \cdot x - P_p \cdot y\right)$$

$$N_B = K_{m,B}\left(P_{f,B} - P_{p,B}\right) = K_{m,B}\left(P_f(1-x) - P_p(1-y)\right)$$

The mol fraction of component A ($y_A = y$) can be expressed as:

$$y = \frac{N_A}{N_A + N_B} = \frac{K_{mA}\left(P_f\, x - P_p\, y\right)}{K_{mA}\left(P_f\, x - P_p\, y\right) + K_{mB}\left(P_f\,(1-x) - P_p\,(1-y)\right)}$$

Dividing by P_f

$$y = \frac{N_A}{N_A + N_B} = \frac{K_{mA}\, P_f\left(x - \frac{P_p}{P_f}\, y\right)}{K_{mA}\, P_f\left(x - \frac{P_p}{P_f}\, y\right) + K_{mB}\, P_f\left((1-x) - \frac{P_p}{P_f}\,(1-y)\right)}$$

Let's define **Pressure Ratio (γ)** as: $\gamma = \frac{P_{permeate}}{P_{feed}}$ $\quad 0 < \gamma < 1$

$$y = \frac{N_A}{N_A + N_B} = \frac{K_{mA}\, P_f (x - \gamma\, y)}{K_{mA}\, P_f (x - \gamma\, y) + K_{mB}\, P_f[(1-x) - \gamma\,(1-y)]}$$

Divide by $K_{m,A} \cdot P_f$

$$y = \frac{(x - \gamma\, y)}{(x - \gamma\, y) + \frac{K_{mB}}{K_{mA}}[(1-x) - \gamma\,(1-y)]}$$

If one defines **selectivity** α as the ratio of permeabilities,

$$y = \frac{(x - \gamma\, y)}{(x - \gamma\, y) + \frac{(1 - x - \gamma + \gamma\, y)}{\alpha}} \qquad (4.5)$$

This can be rearranged to:

$$y = \frac{\alpha\, x - \alpha\, \gamma\, y}{\alpha\, x - \alpha\, \gamma\, y + 1 - x - \gamma + \gamma\, y}$$

$$\alpha\, x\, y - \alpha\, \gamma\, y^2 + y - x\, y - \gamma\, y + \gamma y^2 - \alpha\, x + \alpha\, \gamma\, y = 0$$

$$\gamma(1-\alpha)y^2 + (\alpha x + 1 - x - \gamma + \alpha\gamma)y - \alpha x = 0$$

$$(1-\alpha)y^2 + \left(\frac{1}{\gamma} + \frac{x}{\gamma}(\alpha-1) - 1 + \alpha\right)y - \frac{\alpha}{\gamma}x = 0$$

$$(\alpha-1)y^2 + \left(1 - \alpha - \frac{1}{\gamma} - \frac{x}{\gamma}(\alpha-1)\right)y + \frac{\alpha}{\gamma}x = 0 \qquad (4.6)$$

$y = f(\alpha, \gamma, x)$

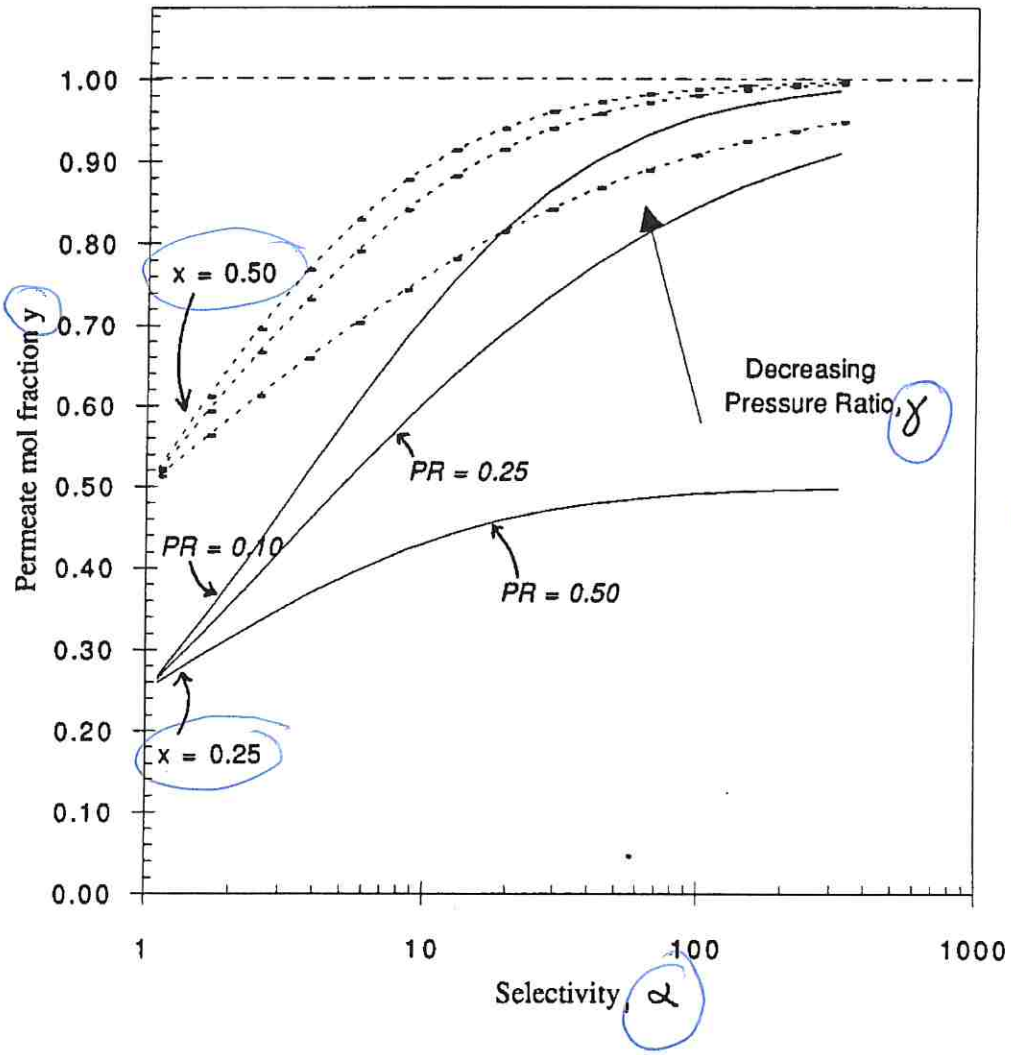
i.e. a quadratic in y. (Remember to check which is the required root i.e. $0 < y < 1$)
Hence knowing x, we can calculate y for a binary mixture.

N.B. as $\gamma \rightarrow 0$, we have the maximum driving force for separation, $P_f >> P_p$ and hence by rearranging Eqn 4.5:

$$y = \frac{\alpha \cdot x}{1 + (\alpha - 1) \cdot x} \qquad (4.7)$$

and this is exactly analogous to the result in equilibrium staged processes, where α is the relative volatility of the binary species; here α is the ratio of permeabilities of the two species.

The general results for y = f (x, α, γ) look like the following. *Note the chart shows results for two values of x (0.25 and 0.5) at three different values of pressure ratio (PR) γ (0.1, 0.25 and 0.5)*



$y = f(\alpha, \gamma, x)$

Notes:

i. When $\alpha = 1$ $y = x$

ii. As $\alpha \uparrow$ $y \to 1$ (unless)

iii. y might approach an asymptote < 1 because equilibrium.

Remember the molar flux can be expressed as:

$$N_A = K_{m,A}(P_{f,A} - P_{p,A})$$

$$N_A = K_{mA} P_f \, (x - \gamma\, y) \qquad (4.8)$$

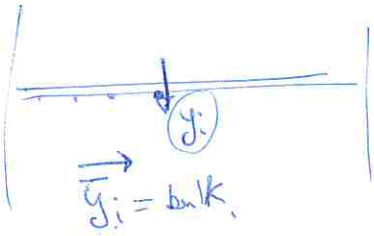
driving force

i.e. when $x = \gamma\, y$, no further separation can occur and y asymptotes to a constant value as α increases.

Knowing the desired volumetric flux, one can calculate the area required for separation:

$$Q_{(2),A} = A \cdot J_A = \frac{A \cdot N_A}{P_{m,A}}$$

iv. For *multicomponent systems* (i.e. >2 species), we cannot obtain an analytical result for the permeate composition and a trial and error iterative method has to be used. Like other multi-component problems, it is inappropriate to attempt to specify too many compositions in the two outlet streams (permeate and retentate) and it is often convenient to specify the recovery of *one key component* and then determine the recoveries of the non-keys that the required design delivers.



4.8.1 CALCULATION OF MEMBRANE AREA

Estimated approached

Let's examine more closely the overall situation:

- The derivations shown in §4.4.1 were made at one (typical) point along the membrane surface and referred to the local value of the concentration of the species in the permeate, y, emerging from the surface of the membrane *at this point*. This is the local or *interfacial concentration* y_i.
- However the average bulk concentration of the species at any point in the *bulk* gas along the permeate side of the membrane will be different to this one local interfacial value and is denoted as $\overline{y_i}$.
- On the assumption of plug flow along the feed side and permeate side of the membrane, the values of x, y_i and $\overline{y_i}$ change continuously along the length of the membrane (from inlet to outlet), thus complicating the analysis.

These additional points are clearly illustrated with a worked example.

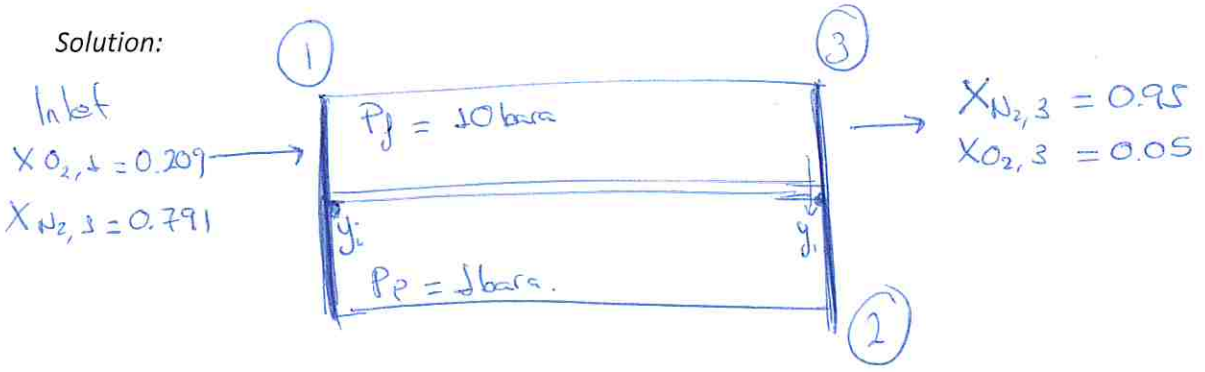
Worked example 4.1: air separation

An air separation membrane has an O_2/N_2 selectivity, α, of 5 and the permeability of the membrane to oxygen is $1.7 \cdot 10^{-5}$ $Nm^3/(m^2\ s\ bar)$. The device operates in a *co-current flow mode* and the retentate contains 95 vol.% nitrogen. [Nm^3 refers to volume measured at *normal* conditions of 1 bara and 25°C]

i. Calculate the permeate composition and the fraction of feed obtained as permeate if $P_{feed\text{-}side}$ = 10 bara and $P_{permeate\text{-}side}$ = 1 bara.

ii. Estimate the area of the membrane needed to treat a feed of 1 Nm^3/s.

Solution:



Pressure ratio : $\gamma = \frac{P_{perm}}{P_{feed}} = \frac{1}{10} = 0.1$ $\alpha = 5$ $k_{m,O_2} = 1.7 \cdot 10^{-5}\ m^3/m^2.s.bar$

Let x = mol raction of oxygen

At inlet, x_1 = 0.209 (feed to the unit is air)

Retentate outlet (i.e. the feed side of the membrane), x_3 = 0.05 (i.e. 95% nitrogen)

i. Find local values of yi using Eqn 4.6, namely:

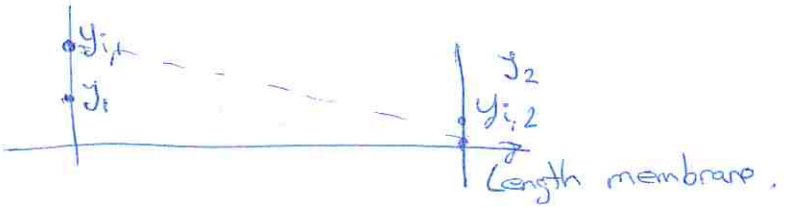
$y = f(\alpha, \gamma, x)$

$$(\alpha - 1)y^2 + \left(1 - \alpha - \frac{1}{\gamma} - \frac{x}{\gamma}(\alpha - 1)\right)y + \frac{\alpha}{\gamma}x = 0 \qquad (4.6)$$

However, the value of y_i depends on the local value of x and hence will change along the length of the membrane.

At inlet: $4y_i^2 - 22.36y_i + 10.45 = 0$ x_i = 0.209, y_i = 0.515

At outlet $4y_i^2 - 16y_i + 2.5 = 0$ x_i = 0.005, y_i = 0.163



Note that at the outlet the mol fraction of oxygen coming through the membrane is less than that in the air supplied at the inlet of the unit.

At this stage, for simplicity, we will assume that an arithmetic average can be used to calculate $\overline{y_i}$ *(but this would only be true if the flux remains constant along the membrane).*

$$\overline{y_i} = \frac{0.515+0.163}{2} = 0.339$$

Hence, as expected, the permeate has been enriched in oxygen.

Material Balance on oxygen over the unit (performed in Nm^3):

$Q_1 = Q_2 + Q_3$

Feed = Permeate + Retentate

$Q = 1\ m^3/s$

$Q_1 \cdot x_{O_2,1} = Q_2 \cdot y_{O_2,2} + Q_3 \cdot x_{O_2,3}$

$1 \cdot 0.209 = Q_2 \cdot 0.339 + (1 - Q_2) \cdot 0.05$

This yields that Q_2 = 0.55 Nm^3/s i.e. the fraction of the total volume of feed that appears as oxygen-enriched permeate is 55%

ii. Now we need to calculate the surface area of the membrane (Basis: 1 Nm^3/s feed)

We need to find the average value of volumetric flux of oxygen across the membrane.

- This flux varies with position along the membrane.
- Ignore film mass transfer effects and assume feed and permeate sides of membrane are at different but constant pressures.

Note: the given value of the permeability of the membrane (units = $Nm^3/(m^2\ s\ bar)$) is a volumetric flux (J_{Ox}) divided by a pressure difference driving force (in bar).

$$J_{ox} = K_{m,ox}(P_f x - P_p y_i) = K_{m,ox} P_f (x - \gamma y_i)$$



At inlet: $J_{ox} = 1.7 \cdot 10^{-5} \cdot 10 \cdot (0.209 - 0.1 \cdot 0.515) = 2.68 \cdot 10^{-5}\ Nm^3 m^{-2} s^{-1}$

At outlet $J_{ox} = 1.7 \cdot 10^{-5} \cdot 10 \cdot (0.05 - 0.1 \cdot 0.163) = 5.73 \cdot 10^{-6}\ Nm^3 m^{-2} s^{-1}$

$$Flowrate\ through\ membrane = Flux \cdot Area\ of\ membrane$$

Area: $$A = \frac{Q_{2,ox}}{J_{ox}} = \frac{Q_2 \bar{y}_i}{J_{ox}} = \frac{0.55 \cdot 0.339}{J_{ox}}$$

But, $\frac{\text{Flux at inlet}}{\text{Flux at outlet}} = 4.68$, so what type of *average flux* should best be used?

Calculate using either:

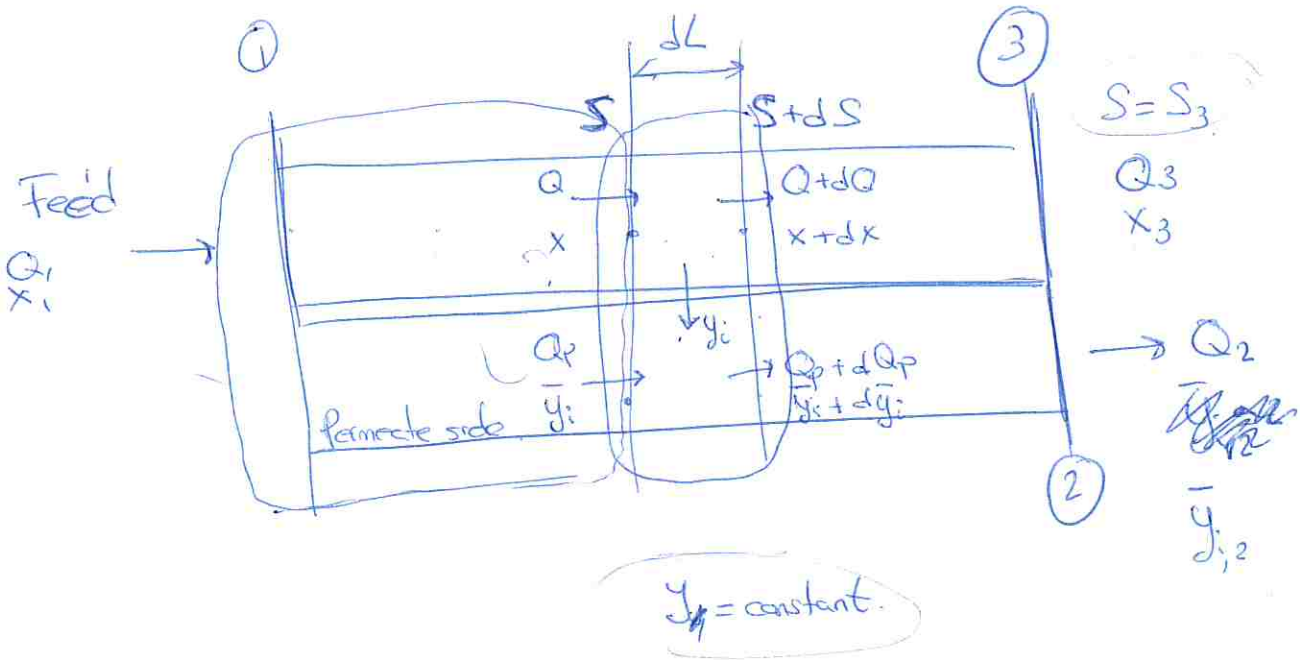
i. Arithmetic Mean Flux = $1.625 \cdot 10^{-5}$ $Nm^3/m^2\,s$ → **Area = 11,470 m^2**

ii. By analogy with previous heat and mass transfer systems involving countercurrent flow, Log Mean Flux = $1.365 \cdot 10^{-5}$ $Nm^3/m^2\,s$ → **Area = 13,650 m^2**

It can be seen in §4.4.2, the flux decreased significantly between the inlet and the outlet of the unit. Arithmetic and log mean values were used to estimate the required membrane area. How accurate are these approaches?

Rigorous approach

Differential analysis for co-current flow:



Assumptions:

- Gas separation takes place, ignoring the concentration-polarisation effect, etc.
- There are no film mass transfer resistance effects
- Assume constant membrane permeabilities and constant pressures on each side of the membrane
- Fluxes of each component (and hence the separation achieved) are f (x, y, α, y_i) and ≠ f($\overline{y_i}$ in bulk permeate).

Let's define the recovery, *S, as:* $S = \frac{Q_p}{Q_1}$

So:

$$dS = \frac{dQ_p}{Q_1} \qquad dQ_p = Q_1 \cdot dS$$

At the entrance of differential volume:

$$Q = Q_1 - Q_p = Q_1 - \frac{Q_1}{Q_1} Q_p$$

$Q = f(S)$ → $Q = Q_1(1 - S)$

At the exit of the differential volume:

$$Q + dQ = Q_1 - (Q_p + dQ_p) = Q_1 - (Q_1 S + Q_1 dS)$$

$Q = f(S)$ → $Q + dQ = Q_1(1 - (S + dS))$

Overall molar balance of solute (at the entrance of the differential volume):

$$Q_1 \cdot x_1 = Q \cdot x + Q_p \cdot \overline{y_i}$$

$Q = Q_1(1-S)$, $Q_p = Q_1 \cdot S$

$$Q_1 \cdot x_1 = Q_1(1 - S) \cdot x + Q_1 S \cdot \overline{y_i}$$

$$\overline{y_i} = \frac{Q_1 \cdot x_1 - Q_1(1 - S) \cdot x}{Q_1 S}$$

$\overline{y_i} = f(x_{inlet}, x, S)$

$$\overline{y_i} = \frac{x_1 - (1 - S) \cdot x}{S} \qquad (4.9)$$

Molar balance of solute over the differential element:

$$Q\,x = (Q + \mathrm{d}Q)\,(x + \mathrm{d}x) + \mathrm{d}Q_p\,y_i$$

$$Q_1\,(1 - S)\,x = Q_1\,(1 - (S + \mathrm{d}S))\,(x + \mathrm{d}x) + Q_1\,\mathrm{d}S\,y_i$$

$$(1 - S)\,x = (x + dx) - (S + dS)(x + dx) + y_i dS$$

$$-x + Sx + x + dx - xS - xdS - Sdx - dSdx + y_i dS = 0$$

$$(1 - S)\,\mathrm{d}x - x\,\mathrm{d}S + y_i\,\mathrm{d}S = 0$$

$$\frac{dS}{(1 - S)} = \frac{dx}{(x - y_i)} \qquad (4.10)$$

For co-current flow over the whole unit, then:

$$\int_{S=0}^{S=S_3} \frac{\mathrm{d}S}{1 - S} = \ln\left(\frac{1}{1 - S_3}\right) = \int_{x_1}^{x_3} \frac{\mathrm{d}x}{x - y_i} \qquad (4.11)$$

For the Integration of Equation 4.11, one should notice that y_i is a complicated function of x (Equation 4.6 requires solution of a quadratic) and hence the integral has to be evaluated numerically.

The solution technique requires the following steps:

(i) Specify a series of values of x between x_1 and x_3

(ii) Calculate the values of y_i for each value of x (Using Equation 4.6)

(iii) Numerically integrate rhs of Equation 4.11 e.g. by trapezium rule and find S.

(iv) Use Equationn 4.9 to find $\bar{y}$.

How to calculate the required membrane area:

- Examine the flow of the component through the differential element of membrane:

$$dQ_p\,y_i = (Q_1\,dS)\,y_i = J_A\,dA \qquad (4.12)$$

where J_A is the local value of the flux of the component concerned.

- Thus, the required membrane area is:

$$A = \int_{inlet}^{outlet} dA = Q_1 \int_0^S \frac{y_i \, dS}{J_A}$$

Again a numerical integration technique is required.

These techniques can be illustrated with a worked example.

Worked Example 4.2

Find area of membrane for the gas permeator in Example 4.1, using numerical integration rather than average fluxes.

Step along the membrane using *k* steps. In this case, e.g. $\Delta x = 0.01$ except for the first step where it is convenient to take $\Delta x = 0.009$.

For each interval:

i. Calculate the values of y_i at the boundaries of the interval using the appropriate values of x in Equation 4.6

Illustration of the calculation for the first increment (with boundaries at *a* and *b*):

$x_a = 0.209$ $x_b = 0.200$ $\Delta x = -0.009$

$y_{i,a} = 0.515$ $y_{i,b} = 0.500$ (using Eqn 4.6)

ii. Calculate the average value of $\frac{\Delta x}{(x-y_i)}$ over the interval and use this to find $S(x)$ at the boundary of the interval using Equation 4.11:

$$\frac{\Delta x}{(x - y_i)} = 0.030$$

$$-\ln(1-S(x)) = \int_{x_1}^{x} \frac{dx}{x-y_i} = \sum_k \frac{\Delta x}{(x-y_i)}$$

Calculate S at x = 0.200:

$$-\ln(1-S(0.20)) = \frac{x_b - x_a}{\left(\left(\frac{x_b + x_a}{2}\right) - \left(\frac{y_{i,b} + y_{i,a}}{2}\right)\right)} = \frac{-0.009}{-0.303} = 0.0297$$

X at dL

$$S(0.20) = 0.029$$

Note: Use average values of both x and y_i within the increment (between *a* and *b*). As this is the first interval, there is only one entry in the summation. The number of entries in the summation will increase with the number of intervals.

iii. Calculate the value of dS that has been achieved across the interval

$$dS = S(0.20) - S(0.209) = 0.029$$

iv. Calculate the average value of $(x - \gamma\, y_i)$ over the interval and use this to calculate local flux using:

$$J_{ox} = K_{m,ox}(P_f x - P_p y_i) = K_{m,ox} P_f (x - \gamma y_i)$$

$$\overline{J_{ox}} = 1.7 \cdot 10^{-5} \cdot 10 \left(\frac{0.209 + 0.200}{2} - 0.1\frac{0.500 + 0.515}{2}\right) = 2.6 \cdot 10^{-5} \frac{Nm^3}{m^2 s}$$

v. Calculate the average permeate concentration $\overline{y}$ from the inlet up to this point (i.e. x = 0.20) in the membrane, using Equation 4.9

$$\overline{y_i} = \frac{x_1 - x(1-S)}{S} = \frac{0.209 - 0.2(0.971)}{0.0293} = 0.507$$

vi. Calculate dA, the area needed to achieved that amount of component flow given the local value of the component flux using Equation 4.12

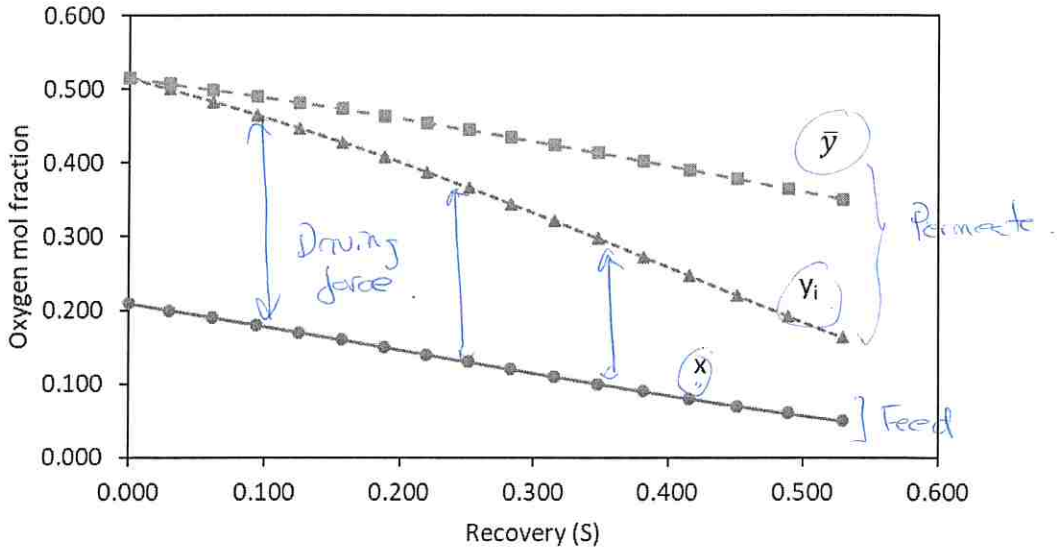
$$dA = \frac{Q_1 \left(\frac{y_{i,a} + y_{i,b}}{2}\right) dS}{J_{ox}} = \frac{1 \cdot 0.507 \cdot 0.0293}{2.6 \cdot 10^{-5}} = 568\ m^2$$

Use a spreadsheet to facilitate the calculations for the other intervals along the membrane and eventually sum all the values of dA to give overall area A.

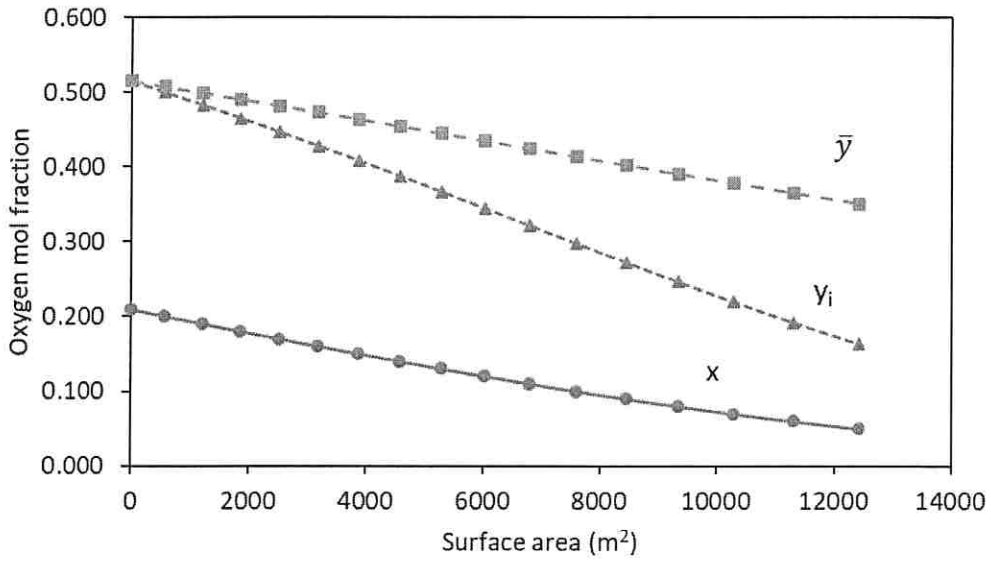
x	yi	$\frac{\Delta x}{x_{av} - y_{i,av}}$	S	$\bar{y}$	dS ($S_{out} - S_{in}$)	Flux Nm^3/m^2s	Area m^2
0.209	0.515	-	0	0.515	-	2.7E-05	0.0
0.200	0.500	0.030	0.029	0.507	0.029	2.6E-05	568
0.190	0.483	0.034	0.061	0.499	0.032	2.4E-05	1206
0.180	0.465	0.035	0.093	0.491	0.032	2.3E-05	1853
0.170	0.447	0.036	0.125	0.482	0.032	2.1E-05	2510
0.160	0.428	0.037	0.157	0.473	0.032	2.0E-05	3179
0.150	0.408	0.038	0.188	0.464	0.031	1.9E-05	3863
0.140	0.388	0.040	0.219	0.454	0.031	1.7E-05	4563
0.130	0.367	0.041	0.251	0.445	0.032	1.6E-05	5283
0.120	0.344	0.043	0.283	0.435	0.032	1.5E-05	6026
0.110	0.321	0.046	0.315	0.424	0.032	1.3E-05	6797
0.100	0.297	0.049	0.348	0.413	0.033	1.2E-05	7601
0.090	0.273	0.053	0.381	0.402	0.033	1.1E-05	8444
0.080	0.247	0.057	0.416	0.390	0.034	9.4E-06	9335
0.070	0.220	0.063	0.451	0.378	0.036	8.2E-06	10285
0.060	0.192	0.071	0.489	0.365	0.038	6.9E-06	11311
0.050	0.163	0.082	0.529	0.351	0.040	5.7E-06	12433

Plotting the results:

Concentration profiles as a f (S):



Concentration profiles as a f (A):



Notes

- The value for the area (12430 m^2) lies between the values previously calculated using by the arithmetic mean flux (11470 m^2) and the logarithmic mean flux (13650 m^2). In this case, the arithmetic mean flux value underestimates the area required and the logarithmic mean flux value overestimates the area required.

- The final recovery of oxygen (0.529) is smaller than the approximate value (0.55) previously calculated and the permeate mol fraction of oxygen (0.351) is larger than the approximate value (0.339). The approximate method slightly overestimates the fraction of the more permeable component (oxygen) that was recovered in the permeate (0.892 *cf* 0.888).

→ Rule of thumb.

- If $\frac{\text{Flux at inlet}}{\text{Flux at exit}} < 2$, then a simple arithmetic average gives a reasonable estimate of the design parameters.

→ 4.68

— Q3 —

4.9 MEMBRANE SEPARATIONS IN LIQUID SYSTEMS

Consider diffusion through the pores of a membrane from a fluid feed to a sweep fluid when identical total pressures but different component concentrations exist on both sides of the membrane. Bulk flow through the membrane due to a pressure difference does not occur, but if species diffuse at different rates, a separation can be achieved.

Driving force = Δ concentrations

There are some important differences between the analysis of gas and liquid membrane separations.

- With liquid separations, *concentrations* rather than mol fractions are used
- The solvent (V) is usually present in very large excess and has the largest permeability as it is the smallest molecule present. Hence:

$$N_V >> N_j \text{ for all } j \neq V$$

Almost all situations can be considered as two-component problems involving the solvent (V) and one solute (B).

The concentration of solute B in the permeate is then: $N_V \gg N_j$

$$C_{B,p} = \frac{N_B}{\sum \frac{N_j}{\rho_{m,j}}} \simeq \frac{N_B}{\frac{N_V}{\rho_{m,V}}}$$

$$N = \frac{Q_p \cdot \rho_m}{A} \qquad \frac{N}{\rho_m} = \frac{Q_p}{A} = J_V$$

Hence:

$$C_{B,p} \simeq \frac{N_B}{J_V}$$

where J_V is the volume flux of solvent (m^3/m^2s).

The molar flux of solute B can be written in terms of a concentration difference driving force:

$$N_B = K_{O,B}\left(C_{B,f} - C_{B,p}\right)$$

$K_{O,B}$: overall membrane permeability

$\left(C_{B,f} - C_{B,p}\right)$: driving force.

Combining both expressions together:

$$K_{O,B}(C_{B,f} - C_{B,p}) = J_V\, C_{B,p}$$

$$K_{O,B}C_{B,f} - K_{O,B}C_{B,p} = J_V\, C_{B,p}$$

$$C_{B,p}(J_v + K_{O,B}) = C_{B,f}\, K_{O,B}$$

$$C_{B,p} = C_{B,f}\left[\frac{K_{O,B}}{K_{O,B} + J_V}\right] \qquad (4.13)$$

Retention, R_B, can then be expressed as:

$$R_B = \frac{C_{B,f} - C_{B,p}}{C_{B,f}} \equiv 1 - \frac{C_{B,p}}{C_{B,f}} = 1 - \frac{K_{O,B}}{K_{O,B} + J_V} = \frac{J_V}{K_{O,B} + J_V}$$

Hence the retention of the solute R_B is determined both by the membrane's overall permeability and the magnitude of the solvent flux.

It is also common to write the solvent volume flux as:

$$J_V = K_{V,O}\,\Delta P \qquad (4.14)$$

where $K_{V,O}$ is an appropriate overall membrane permeability of solvent *for use with pressure driving forces* (typical units = $m^3/(m^2\ s\ bar)$).

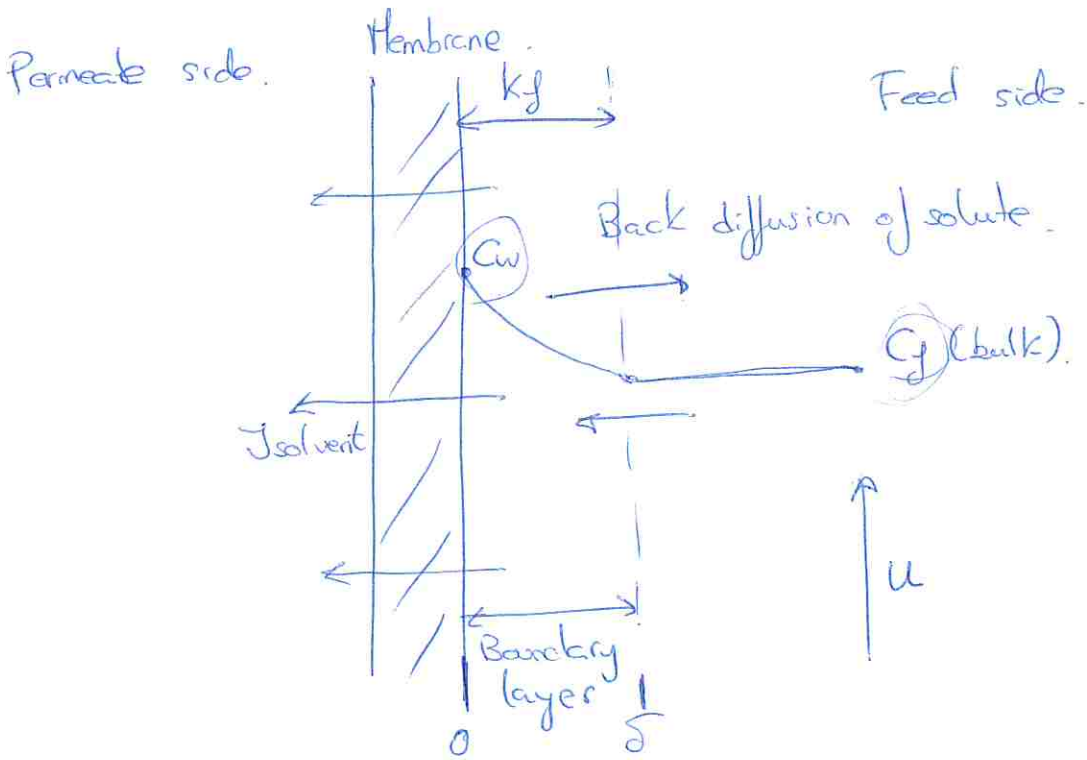
4.9.1 Concentration Polarisation (CP)

This is a phenomenon in membrane separations whereby the flux through the membrane is controlled by the film mass transfer resistance on the feed-side (characterised by the mass transfer coefficient k_f) rather than the resistance of the membrane itself K_m. *The effect is most noticeable in liquid separations*.

Process	Influence of CP	Origin
Hyperfiltration	moderate	large *kf*
Ultrafiltration	strong	small *kf* or large flux
Microfiltration	strong	small *kf* or large flux
Gas Separation	(very) low	large *kf* or small flux

Origins of concentration polarisation:

A solute that is selectively retained by the membrane will accumulate at the membrane surface and generates a mass transport resistance for solute and solvent transfer. The flux of solvent + solute towards the membrane convects solute to the membrane surface where it becomes stranded in the stagnant boundary layer (thickness δ) at the wall (a bit like the build-up of a filter cake in conventional filtration). In the steady state, this is balanced by the back diffusion of the solute down the developing concentration gradient.



Let C = the concentration of the solute (B). In the steady state, in direction z.

$$N_B = -J_V C_p = -J_V C - D_B \frac{dC}{dz}$$ → Fick's law.

$$-J_V C_p + J_V C = -D_B \frac{dC}{dz} \qquad J_V (C - C_p) = -D_B \frac{dC}{dz}$$

$$\int_0^{\delta} J_V \, dz = \int_{C_w}^{C_f} -D_B \frac{dC}{(C - C_p)}$$

$$J_V \delta = D_B \ln\left(\frac{C_w - C_p}{C_f - C_p}\right)$$

$K_f = \frac{D_B}{\delta}$

$$J_V = k_f \ln\left(\frac{C_w - C_p}{C_f - C_p}\right) \quad (4.15)$$

Concentration polarisation equation.

where k_f is a film mass transfer coefficient across the stagnant film.

Hence, the solvent flux depends critically on the profile of solute concentration on the feed side of the membrane.

The higher the value of J_V, the higher the value of C_w will become unless k_f can be increased, e.g. by increasing the rate of bulk tangential flow across the membrane surface (perpendicular to the flux through the membrane) or by trying to induce more turbulence in the system.

k_f can be predicted from mass transfer correlations. For example, for turbulent flow of fluid inside a tubular membrane of internal diameter d:

$$\mathrm{Sh} = \frac{k_f d}{D_B} = 0.023 \mathrm{Re}^{0.8} \mathrm{Sc}^{0.4}$$

High values of C_w have a number of consequences:

- It increases the local osmotic pressure (see later) and thus decreases solute flux
- For membranes working on the "solution-diffusion" principle, the value of the rejection R for the solute will be decreased as the flux of B through the membrane $N_B \propto (C_w - C_p)$ will be higher.
- Increased risk of membrane fouling.

4.9.2 Gel effect and limiting flux

The concentration polarisation equation 4.15 suggests that C_w should increase indefinitely with further increases in flux J_V. This does not occur in practice due to two effects;

- The formation of a solute "gel" – the *gel effect*
- Increased osmotic pressure (due to increased C_w) actually reduces J_V

For a macromolecular solute (e.g. a protein), as C_w increases, its concentration may approach its *gel concentration* C_G at which it forms a gel (rather than a concentrated solution).

This sort of a gel is a "structured liquid" of near infinite viscosity (rather like a solid) which prevents further passage of solute through the membrane and itself has a significant resistance to the flux of solvent.

When "gelling" occurs: $C_w = C_G$.

and hence a **limiting solvent flux** J_V^* is reached, given by substitution into Eqn. 4.15:

$$J_v^* = K_{V,0}\Delta P = k_f \ln\left(\frac{C_G}{C_f}\right) \qquad (4.16)$$

$C_p = 0$

A further increase in pressure drop across the membrane at this stage will cause a momentary increase in J_V, but the extra flux will cause a thicker layer of gel to form which will decrease $K_{V,O}$ and the flux will decline back down to J_V. Hence a limiting solvent flux is encountered.

Worked example 4.3

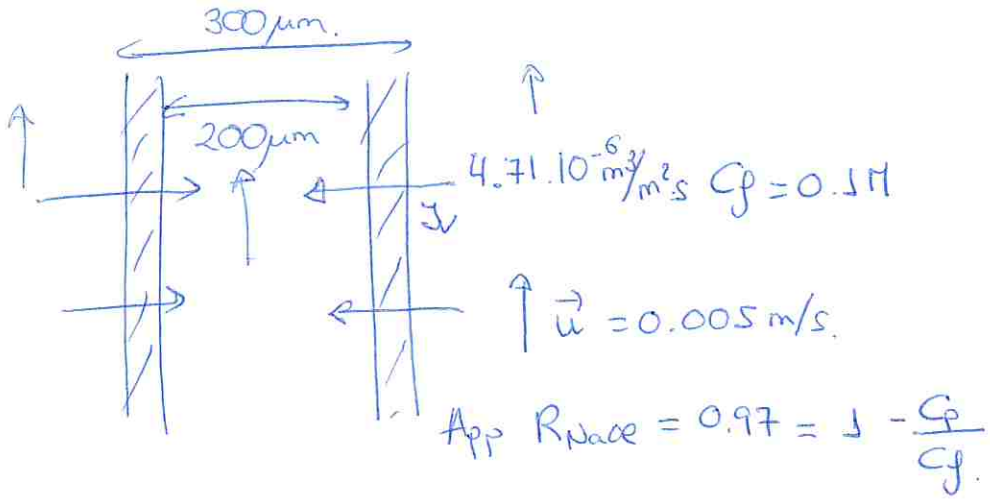
A membrane unit for desalination containing hollow fibres, with d_o = 300 µm and d_i = 200 µm gives a water flux of $4.71 \cdot 10^{-6}$ m^3/m^2 s with a feed containing 0.1 M NaCl at 20°C. The feed solution flows normal to the outside of the fibres at 0.005 m/s and the apparent solute retention ratio (rejection coefficient) (R_{NaCl}) is 0.97. Is concentration polarisation likely to be significant?

$C_w > C_f$

Assume that 0.1 M NaCl has the same physical properties as water and that $D_{NaCl} = 1.6 \cdot 10^{-9}$ m^2/s.

For flow normal to a cylinder use: $\mathrm{Sh} = 0.61 \mathrm{Re}^{1/2} \mathrm{Sc}^{1/3}$

Solution:



Let's calculate the mass transfer coefficient, k_f on the feed side (outside of the hollow fibre):

$$Re = \frac{u\, d\, \rho}{\mu} = \frac{0.005\ \frac{m}{s}\ 300 \cdot 10^{-6} m\ 1000 \frac{kg}{m^3}}{0.001 \frac{kg}{m\ s}} = 1.5$$

$$Sc = \frac{\mu}{\rho\, D} = \frac{0.001 \frac{kg}{m\ s}}{1000 \frac{kg}{m^3}\ 1.6 \cdot 10^{-9} \frac{m^2}{s}} = 625$$

$$Sh = 0.61\, Re^{1/2} Sc^{1/3} = 0.61 \cdot 1.5^{0.5} \cdot 625^{0.33} = 6.37$$

$$Sh = \frac{k_f\, d}{D_B} \qquad k_f = \frac{Sh\, D_B}{d} = \frac{6.37 \cdot 1.6 \cdot 10^{-9}\,\frac{m^2}{s}}{300 \cdot 10^{-6} m} = 3.4 \cdot 10^{-5}\,\frac{m}{s}$$

The concentration of salt in the permeate (inside the hollow fibre) can be calculated as:

$$R_{NaCl} = 1 - \frac{C_p}{C_f} \qquad \frac{C_p}{C_f} = 1 - R_{NaCl}$$

$$C_p = C_f(1 - R) = 0.1\ M\ (1 - 0.97) = 0.003\ M$$

We can also calculate the concentration of salt at the membrane wall (feed side) by using the concentration polarisation equation (4.15):

$$J_V = k_f \ln\left(\frac{C_w - C_p}{C_f - C_p}\right) \qquad (4.15)$$

Which rearranges to:

$$\left(\frac{C_w - C_p}{C_f - C_p}\right) = exp\left(\frac{J_V}{k_f}\right)$$

$$C_w - C_p = (C_f - C_p)exp\left(\frac{J_V}{k_f}\right) \qquad C_w = C_p + (C_f - C_p)\exp\left(\frac{J_v}{k_f}\right)$$

$$C_w = 0.003 + 0.097\ \exp\left(\frac{4.71 \cdot 10^{-6}}{3.4 \cdot 10^{-5}}\right) = 0.114\ \mathrm{M}$$

Is this significant?

Rule of thumb →

$$\frac{C_w - C_f}{C_f} = \frac{0.114 - 0.1}{0.1} = 0.14$$

and the effect is considered to be just about **significant** (> 0.1).

True solute rejection coefficient:

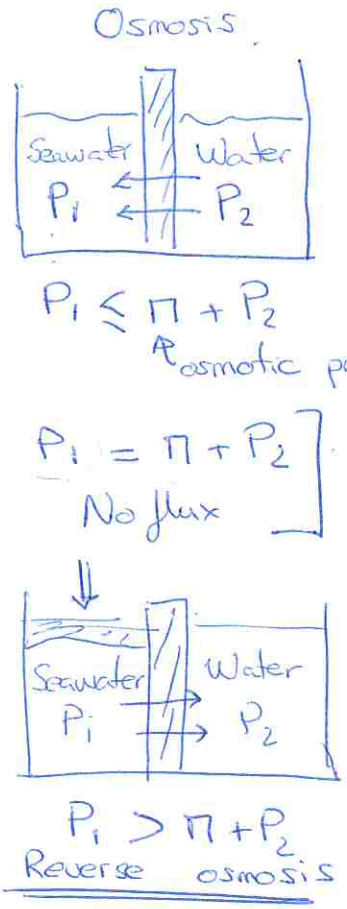
$$R_{NaCl,true} = \frac{C_w - C_p}{C_w} = \frac{0.114 - 0.003}{0.114} = 0.973$$

i.e. not much altered from the apparent value of R_{NaCl}.

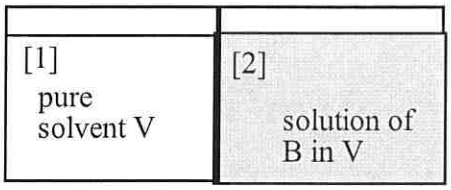
4.9.3 Osmotic Pressure and Reverse Osmosis

An understanding of **osmotic pressure** is necessary for the analysis of ultrafiltration and hyperfiltration (reverse osmosis) processes.

- When miscible solutions containing different concentrations of solute are separated by a membrane that is permeable to the solvent but nearly impermeable to the solute, diffusion of the solvent occurs from the less concentrated to the more concentrated solution and is called **osmosis**.
- This transfer can be stopped if the pressure of the concentrated solution is increased until the activity of the solvent is the same on both sides of the membrane.
- If a pressure difference greater than the osmotic pressure difference between both sides of the membrane is applied, solvent will move from the concentrated solution through the membrane into the more dilute solution, i.e. **reverse osmosis**. This is what occurs in ultra-filtration and hyperfiltration and the latter technique is sometimes referred to as reverse osmosis.



Osmosis (from the Greek 'to push') is thus the tendency for a pure solvent (V), to enter a solution of solute B separated from it by a semi-permeable membrane.



membrane permeable to V, not B

The concentration of solvent V in phase [1] is higher than that in phase [2]. If the pressure in each phase is the same, there will be a difference in the chemical potential of V in the two phases. The general expression for the chemical potential of V in a phase is given by:

$$\mu_V(P) = \mu_V^o + RT \ln a_V + \int_{P^o}^{P} V_{m,V} dP$$

Where: a_V is the activity of V in the solution,

$V_{m,V}$ the molar volume of V and P the pressure.

If $V_{m,V}$ is approximately constant (since it is a liquid):

Phase 1: $\mu_{V,1} = \mu_V^o + V_{m,V}(P_1 - P^o)$; since $a_{V,1} = 1$

Phase 2 $\mu_{V,2} = \mu_V^o + RT \ln a_{V,2} + V_{m,V}(P_2 - P^o)$

As $RT \ln a_{V,2}$ is negative (since $a_{V,2} < 1$) and if $P_1 = P_2$, then $\mu_{V,1} > \mu_{V,2}$

This difference in chemical potentials will promote the diffusion of solvent V across the membrane from [1] in order to dilute the solution [2] (by increasing the amount of solvent in it) and thus increase $\mu_{V,2}$ until equilibrium is established.

This direction of mass transfer is completely the opposite of the one that is sought in ultra- and hyper-filtration, where we want to remove solvent from solute.

This flux due to osmotic pressure can be stopped by applying an additional pressure Π to the solution so that the chemical potentials are equalised:

$$\mu_V^o + \bar{V}_V P_1 = \mu_V^o + RT \ln a_{V,2} + \bar{V}_V (P_1 + \Pi)$$

Giving:

$$\Pi = \frac{-RT \ln a_{V,2}}{V_{m,V}} \qquad (4.17)$$

The increment in pressure, Π, is known as the *osmotic pressure* and represents the driving force promoting dilution of the solution by the solvent.

This pressure difference due to osmotic pressure must be exceeded in ultra– and hyper–filtration to drive solvent molecules out of the solution across a membrane into the solvent phase (permeate).

Effect of osmotic pressure on membrane flux

The solvent flux transport equation (Equation 4.14) has to be modified to allow for the reduction in the pressure driving force due to the presence of osmotic pressure effects on both sides of the membrane. On the feed/retentate side, these are dictated by the concentration of solute at the surface of the membrane (the wall).

$$J_v = K_v\left[\Delta P - (\Pi_w - \Pi_p)\right] \qquad (4.18)$$

- The flux can thus be substantially reduced as a result of large values of Π_w arising from concentration polarisation effects (i.e. high values of C_w).
- For membranes where $R_B = 1$, i.e. the solute cannot pass through the membrane, $\Pi_p = 0$ (there is no osmotic pressure on the permeate side of the membrane as the liquid here is pure solvent).

Van't Hoff Equation

We want to be able to estimate the osmotic pressure as a function of the concentration of solute B.

Consider a dilute solution containing solvent V and solute B. The activity of V in solution (i.e. phase 2 above) is given by:

$$a_{V,2} = \gamma_{V,2}\, x_{V,2} \approx 1 \cdot (1 - x_{B,2}) = (1 - x_{B,2})$$

If this phase is in equilibrium with pure solvent (phase 1), there must be an osmotic pressure given by Eqn 4.17:

$$\Pi = -\frac{RT}{V_{m,V}} \ln(1 - x_{B,2})$$

Simplifying, using the first term of a series expansion for ln(1 – x) and knowing $n_V >> n_B$:

$$\Pi \approx \frac{RT}{V_{m,V}} x_{B,2} = \frac{RT}{V_{m,V}} \frac{n_B}{n_B + n_V} \approx \frac{RT}{V_{m,V}} \frac{n_B}{n_V}$$

Where: *ni* is the number of mols of component *i* in the solution.

This gives:

$$\Pi \approx \frac{RTn_B}{V}$$

where : V is the total volume of solution.

If B is a substance that on dissolution dissociates to give f_B moles of soluble species per mole, e.g. $f_{NaCl} = 2$, $f_{Sucrose} = 1$, $f_{CaCl2} = 3$

$$\Pi = RTC_B f_B \qquad (4.19)$$

C_B is a molar concentration (M or kmol/m^3). **This is a simple *Van't Hoff Equation.***

However macromolecular species exhibit noticeable non-ideality, and alternative, more complicated empirical expressions for Π have to be used, for example:

$$\Pi = A_1 C_B + A_2 C_B^2 + A_3 C_B^3$$
$$or \qquad (4.20)$$
$$\Pi = \varphi C_B^{\beta}$$

Note the value of C_B (kmol/m^3) for a solution of a large macromolecule is often very small, so in general values of Π for solutions of macromolecules are smaller than those pertaining to solutions of small solutes (e.g. salt).

— Q4 —

The technique of *ultrafiltration* can be used to illustrate the interaction between the effects of concentration polarisation and osmotic pressure.

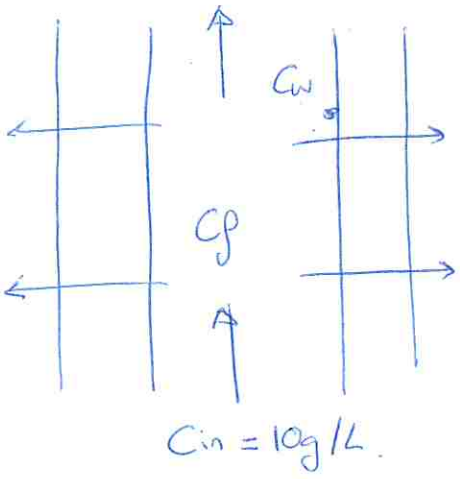
The higher the amount of concentration polarisation, the greater the proportion of the imposed pressure drop across the membrane has to be used to overcome the increased osmotic pressure due to high concentrations of solute at the wall.

Worked example 4.4: ultrafiltration

A tubular ultrafiltration membrane [2 cm diameter, $K_{V,0}$ (water) = 250 L/(m² h bar)] is to be used for concentrating aqueous solutions of 10 g/L cheese whey protein. The diffusion coefficient of whey in H_2O = 4 ·10^{-11} m²/s. The osmotic pressure of a whey solution containing milk proteins at a concentration *C* is given by:

$$\Pi(bar) = 0.0044\ C(g/L) - 1.7 \cdot 10^{-6} C^2 + 7.9 \cdot 10^{-8} C^3$$

Investigate the effect of different operating pressures on the flux of water when the mean flow velocity of whey solution through the inside of the hollow tubes is 1.5 m/s, giving a film mass transfer coefficient k_f = 7.8 ·10^{-6} m/s, and the membrane completely rejects the whey proteins (*R* = 1). The gel limit for whey proteins is C_G = 400 g/L.



$K_{V,0} = 250\ L/(m^2 \cdot h \cdot bar)$

$D_B = 4 \cdot 10^{-11}\ m^2/s$

$u = 1.5\ m/s$

$k_f = 7.8 \cdot 10^{-6}\ m/s$

$R = 1$

$C_G = 400\ g/L$

Note (but not needed for solution):

$$Re = \frac{u\,d\,\rho}{\mu} = \frac{1.5\,\frac{m}{s}\;0.02\,m\;1000\,\frac{kg}{m^3}}{0.001\,\frac{kg}{m\,s}} = 30{,}000$$ (turbulent flow inside tube)

$$Sc = \frac{\mu}{\rho\,D} = \frac{0.001\,\frac{kg}{m\,s}}{1000\,\frac{kg}{m^3}\;4\cdot 10^{-11}\,\frac{m^2}{s}} = 25{,}000$$ (i.e. >> 1, unlike gas systems)

(i) Calculate minimum operating pressure to give $J_V = 0$

Use flux equation in the presence of osmotic pressure Eqn 4.18:

$$J_v = K_{V,O}\left[\Delta P - (\Pi_w - \Pi_p)\right]$$

$J_V = 0$; driving force $= 0$

Calculate osmotic pressures using given correlation:

a. At the wall - In the absence of solvent flux, there is no concentration polarisation and hence $C_f = C_w$. On the feed side, whey concentration is C_f = 10 g/L, and using the given correlation, Π_W= 0.044 bar

b. On the permeate side - Permeate whey concentration C_p = 0 g/L (since R = 1)

Π_p = 0 bar

Hence $\Delta\Pi$ = 0.044 bar and thus the minimum operating driving pressure ΔP for start of successful operation is 0.044 bar, i.e. J_V is negative until this minimum pressure drop is exceeded.

$\Delta P = \Delta\Pi = 0.044 - 0 = 0.044$ bar.

(ii) Investigate the conditions at the gel limit

correlation given.

At the gel limit; $C_w = C_G$ = 400 g/L; Π_G = 6.54 bar

Equation (4.16)

$$J_v^* = k_f \ln\left(\frac{C_G}{C_f}\right)$$

$C_p = 0$

Limiting flux

$$J_V^* = 7.8\cdot 10^{-6}\,\frac{m}{s}\,ln\left(\frac{400\,M}{10\,M}\right) = 2.88\cdot 10^{-5}\,\frac{m}{s} = 103.6\,\frac{L}{m^2\,h}$$

$$\Delta P = \Pi_w + \frac{J_V^*}{K_v} = 6.54\ bar + \frac{103.6 \frac{L}{m^2\ h}}{250 \frac{L}{m^2\ h}} = 6.95\ bar$$

This is the operating pressure at which the limiting flux will be reached. Increasing the pressure yet further will not increase the flux.

(iii) Investigate values of ΔP needed for operation at higher solvent fluxes

For example, let's consider arbitrarily Jv = 0.01 L/m² s <> 36 L/m² h

We need to consider the effects of concentration polarisation, and hence need to evaluate C_w at this flux rate. Remember C_p = 0 since R = 1.

Use Eqn 4.15:

$$J_v = k_f \ln\left(\frac{C_w - C_p}{C_f - C_p}\right) = k_f \ln\left(\frac{C_w}{C_f}\right)$$

$$0.01 \cdot 10^{-3} \frac{m^3}{m^2\ s} = 7.8 \cdot 10^{-6} \frac{m}{s} \ln\left(\frac{C_w}{10}\right)$$

Cw = 36 g/L i.e. 3.6 times greater than in the bulk feed

Estimate Π_w using the correlation given:

$$\Pi(bar) = 0.0044\ C(g/L) - 1.7 \cdot 10^{-6} C^2 + 7.9 \cdot 10^{-8} C^3$$

Π_w = 0.16 bar

Using Equation 4.18:

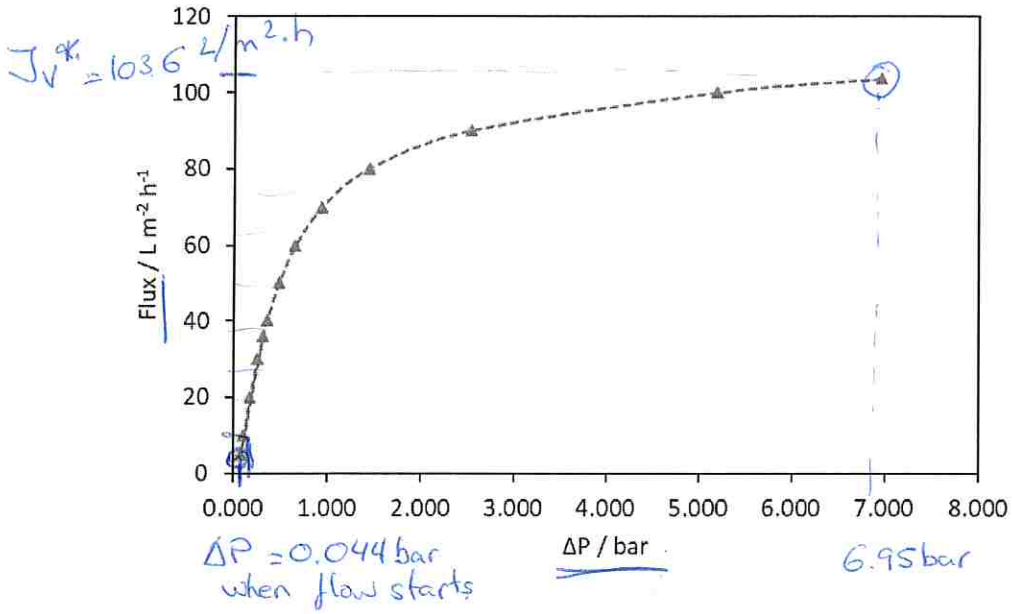
$$J_v = K_{V,O}\left[\Delta P - (\Pi_w - \Pi_p)\right]$$

$$\frac{J_V}{K_V} = \Delta P - \Pi_w$$

$$\Delta P = \Pi_w + \frac{J_V}{K_V} = 0.16 bar + \frac{36\left(\frac{L}{m^2 h}\right)}{250\left(\frac{L}{m^2 h}\right)} = 0.304\ bar$$

- Most of the energy needed for separation at this flux rate (i.e. to create and maintain the operational ΔP) is used to overcome the osmotic pressure caused mainly by concentration polarisation.

(iv) Repeat (iii) for a series of other fluxes between 0 and J^*_V to produce the following graph



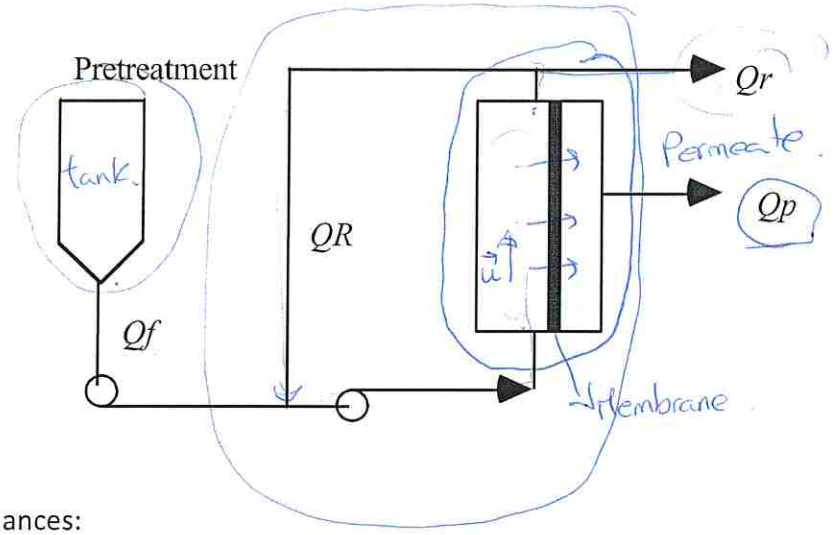
Hence operation at higher values of ΔP does not proportionally increase the observed flux. Unfortunately we cannot get higher and higher fluxes just by increasing the ΔP across the membrane.

4.9.4 Continuous operation

Single pass continuous operation is not practical if concentration polarisation is significant, as high solution velocities are required to control its extent (i.e. by increasing k_f).

- The amount of solvent passing through the membrane during a single pass of feed across the membrane will be very small as the residence time of the liquid in contact with the membrane surface is small (due to the high velocities and high volumetric flow rates of liquids passing through the system).
- The degree of concentration of the retained solute that would be achieved per pass would be low unless large membrane areas are used (uneconomic).

- Continuous operations thus feature a 'feed and bleed' approach, where a recycle loop (with high flow rate Q_R) is used to maintain high velocities through the membrane in a "multi-pass" system.



Volumetric balances:

Overall: $Q_f = Q_r + Q_p$

Around the membrane unit: $Q_R + Q_f = Q_R + Q_r + Q_p$

Under these circumstances: $k_f = f(Q_R + Q_f)$, so Q_f can be reduced to achieve good separation.

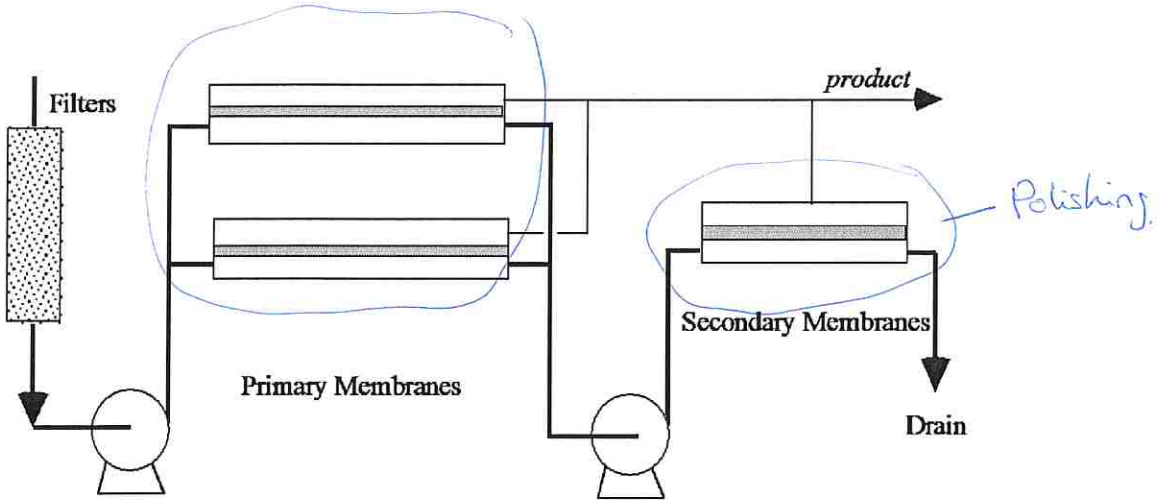
In the absence of recycle: $k_f = f(Q_f)$. Q_f has to be kept high, which leads to a small amount of mass transfer in the single pass. ⇒ very high sa membrane.

4.9.5 Hyperfiltration (reverse osmosis)

Hyperfiltration, also known as reverse osmosis, is a reliable, established technology based on asymmetric or composite membranes (cellulose acetates or polyamides). The most important application of HF is in desalination i.e. the removal of soluble salts from water, where the energy demand is 6-7 kWh/m^3 H_2O product (approximately 23.4 MJ/m^3), or around 50% of the value involved using evaporation.

The driving force for separation by hyperfiltration is fluid pressure in the liquid which is used to overcome membrane resistance and osmotic pressure (hence called "reverse" osmosis). In membrane desalination units, the brine has an associated osmotic pressure which must be exceeded before water will cross the membrane to the almost pure water permeate phase. The osmotic pressure of brackish water, with ~10 g/L TDS (total dissolved solids) is around 7 bar, while sea water, with ~35 g/L TDS, has an osmotic pressure of around 25 bar. Potable (drinkable) water should contain less than 0.25 g/L of dissolved solids.

Industrial applications often use values of $\Delta P \sim 2\text{–}4\ \Delta\Pi$. A typical arrangement is;



- Staged separations are used (i.e. primary membranes followed by secondary membranes), as the high salt rejection by the membrane (necessary for good quality drinking water) results in low flux across the membrane.

- Single stage operations are limited by the low fluxes that result from concentration polarisation and need high membrane surface areas, so units for producing high volumetric flowrates of water are usually multistage.
- The salt concentration on the feed side of the membrane increases from stage to stage, so that the effects of concentration polarisation are greatest in the later stages.
- Feed streams to the later stages are combined and the increased flow rates result in larger mass transfer coefficients, which help counteract the concentration polarisation.

The solvent flux transport equation is written as:

driving force

$$J_v = K_{V,O}\left[\Delta P - (\Pi_w - \Pi_p)\right] = K_{V,O}\left[\Delta P - \varphi(C_w^{\beta} - C_p^{\beta})\right] \qquad (4.21)$$

Where: $K_{V,O}$ is the permeability of the solvent in the membrane,
and ϕ and β are osmotic pressure constants for the empirical correlation for osmotic pressure (Eqn. 4.20).

There are significant concentration polarisation and osmotic pressure effects.
Consider the relationship between the flux of solvent (V) and a solute (B), where all concentrations and mass transfer parameters refer to B unless indicated;

Solute transport equation:

$$N_B = J_V C_p = K_{O,B}(C_w - C_p) \qquad (4.22)$$

Concentration polarisation (equivalent to equation 4.15):

$$J_V = k_f \ln\left[\frac{C_w - C_p}{C_f - C_p}\right] \qquad (4.23)$$

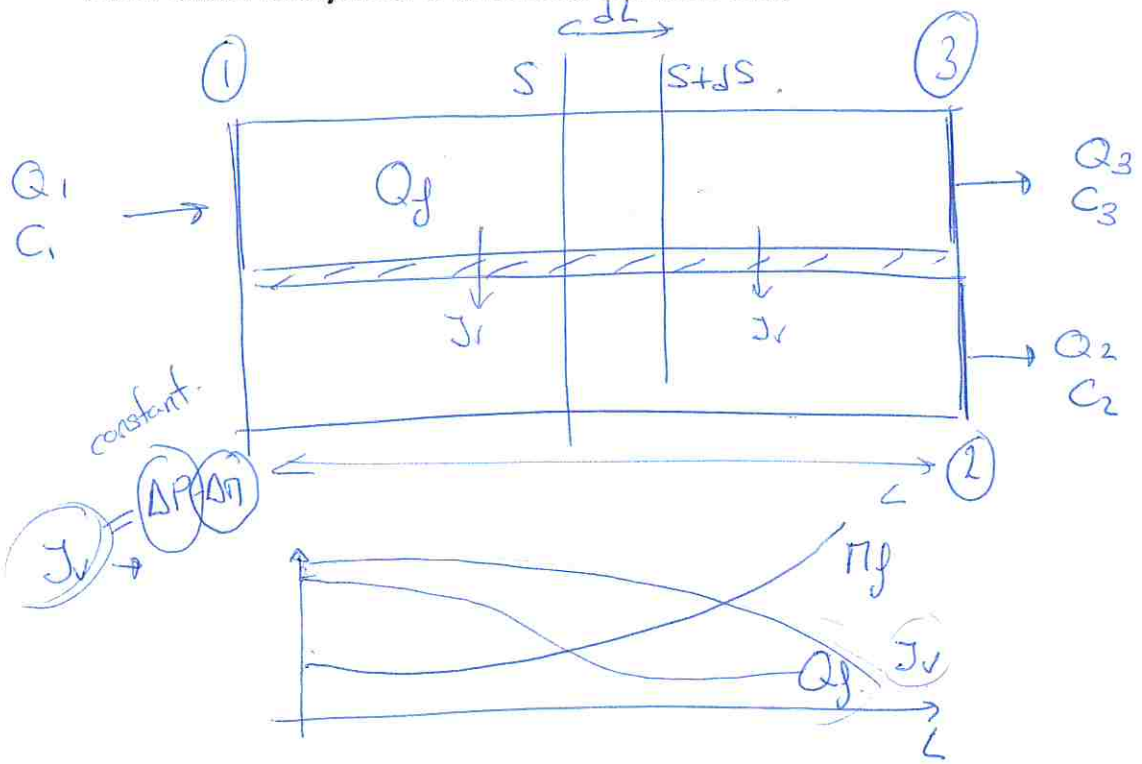
Thus we have three simultaneous equations (Equations 4.21, 4.22 and 4.23) in Jv, Cp and Cw.

The concentration polarisation equation is non-linear so the problem frequently involves a numerical solution approach.

Approximations are often made to simplify the analysis include:

- The osmotic pressure relationship (Eq 4.21) is linear (i.e. the osmotic co-efficient $\beta = 1$)
- concentration polarisation effects are not large
- R_B is constant, termed the **constant salt rejection approximation**

Differential analysis of a Reversed Osmosis Unit



- As recovery S increases along the length of the membrane, Q_f falls and so k_f falls, giving a reduced J_V
- Also, as S increases, C_f increases, giving increased Π_f and further reduction in J_V.

Assumptions:

- R = constant = R_{app}
- Ignore concentration polarisation (as was justified by Worked Example 4.3)

From a molar balance on the element, by direct analogy with Eqn. (4.10) derived for gas separations; here *x* and *y* are mol. fractions of the solute.

$$\int_0^{S_3} \frac{dS}{1-S} = \int_{in}^{out} \frac{dx}{x - y_i}$$

$$\int_0^{S} \frac{dS}{1-S} = \int_{in}^{out} \frac{dC_f}{C_f - C_p} = \int_{C_1}^{C_3} \frac{dC_f}{R_{app} \cdot C_f}$$

Integrating both sides of the equation:

$$\ln\left(\frac{1}{1-S}\right) = \frac{1}{R_{app}} \ln\left(\frac{C_3}{C_1}\right)$$

giving:

$$\frac{C_3}{C_1} = \frac{1}{(1-S)^{R_{app}}} \qquad (4.24)$$

The bulk permeate concentration, C_2, is found from material balances:

$$Q_1 = Q_2 + Q_3 \qquad Q_1 \cdot C_1 = Q_2 \cdot C_2 + Q_3 \cdot C_3$$

$$C_1 = \frac{Q_2}{Q_1} C_2 + \frac{(Q_1 - Q_2)}{Q_1} C_3$$

Since: $S = \frac{Q_2}{Q_1}$, then $C_1 = S\,C_2 + (1-S)C_3$

So:

$$\frac{C_2}{C_1} = \frac{1}{S}\left(1 - \frac{C_3}{C_1}(1-S)\right)$$

Substituting Eqn. 4.24:

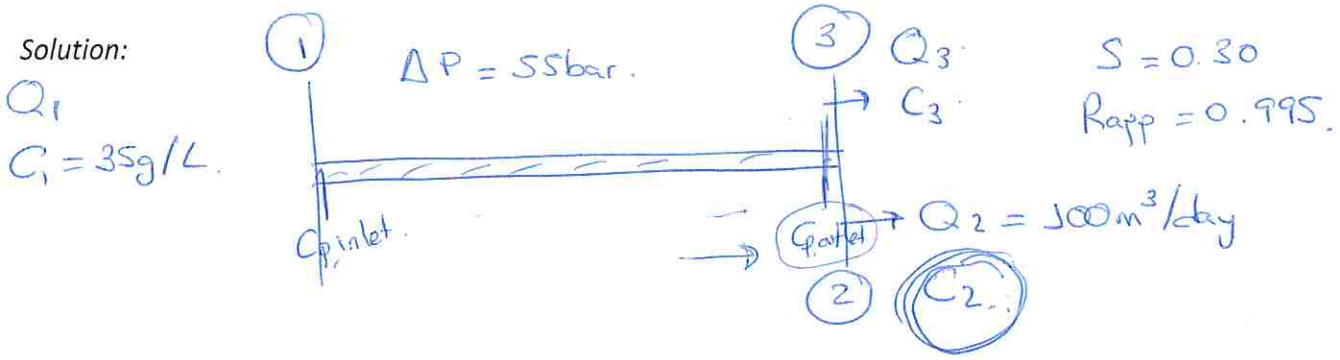
$$\frac{C_2}{C_1} = \frac{1}{S}\left(1 - (1-S)^{1-R_{app}}\right) \qquad (4.25)$$

Worked example 4.5: Hyperfiltration (reverse osmosis)

A membrane desalination process is to produce 100 m^3/day of potable water from sea water containing 35 g/L dissolved salts. The membranes are operated such that the recovery is 30% and have an apparent solute rejection fraction of 0.995. The permeability of water in this case is 0.004 $m^3/(m^2\ h\ bar)$ and the imposed pressure drop across the membrane is 55 bar. The osmotic pressure of salt solutions can be predicted from $\Pi(bar) = 0.82C\ (g/L)$.

How many membrane modules each of area 3 m^2 each are needed?

Solution:



i. Calculate the retentate concentration using Eqn 4.24: $\frac{C_3}{C_1} = \frac{1}{(1-S)^{Rapp}}$

$$C_3 = C_1(1-S)^{-R} = 35\ (0.7)^{-0.005}$$

$$C_3 = 49.9\ g/L$$

f(S, Rapp)

ii. Calculate the permeate concentration using Eqn 4.25:

$$\frac{C_2}{C_1} = \frac{1}{S}(1-(1-S)^{1-R_{app}})$$

$$C_2 = \frac{35}{0.3}(1-(1-0.3)^{0.005}) = 0.208\ \text{g/L}$$

(note: C_2 < 0.25 g/L, so the water produced is indeed potable)

iii. Calculation of the fluxes - require osmotic pressures at inlet and outlet (for an approximate calculation).

$$J_v = K_{V,O}(\Delta P - \Delta\Pi)$$

driving force when affected by osmotic pressure.

Calculate local permeate concentrations using $(C_p = C_f(1-R))$ and then find Π values:

	C_f (g/L)	Π_f (bar)	Cp (g/L)	Π_p (bar)	$\Delta P - \Delta\Pi$ (bar)
Inlet	35.0	28.7	0.175	0.14	26.5
Outlet	49.9	40.9	0.249	0.205	14.3

The ratio of net pressure difference driving forces is just less than 2 so it is appropriate to use the arithmetic average of the pressure difference driving forces to calculate the surface area (as was justified by Worked Example 4.1).

- If concentration polarisation is significant, the true fluxes will be smaller and the ratio of inlet/outlet fluxes will be different. The flux and concentration of solute at the wall are linked by the non-linear concentration polarisation equation so the solution would have to be found by iteration.

Use arithmetic mean of the net driving forces to find:

$$J_v = 0.004 \left(\frac{m^3}{m^2 h\ bar}\right)\left(\frac{26.4 + 14.3}{2}\right)(bar) = 0.081 \left(\frac{m}{h}\right) \approx 1.954 \left(\frac{m}{day}\right)$$

Equate the volumetric production rates for a system with n modules of surface area 3 m^2 each:

$$Q = J_V A$$

$$100 \left(\frac{m^3}{day}\right) = 1.954 \left(\frac{m}{day}\right) \cdot A$$

A = 51.17 m^2 $\qquad$ $modules = \frac{51.17\ m^2}{3\ m^2} \approx 17$

4.9 Membrane Fouling

Fouling is a generic term for phenomena which cause a decline in permeate fluxes over time when all other parameters remain constant. Membrane fouling is the major problem in implementation of membrane technology. It is analogous to heat exchanger fouling and is caused by all sorts of impurities in the feed e.g. oil from marine accidents in the seawater fed to reverse osmosis desalination units.

Consequences of membrane fouling are;

- Loss in flux and capacity
- The need for regular cleaning and its associated costs and downtime
- Changes in the transport properties of the membrane as the membrane deteriorates over time and with each cycle of cleaning

Membranes will periodically need to be renewed and this is an additional appreciable running cost.

Important to install purification stages before the membrane unit in order to reduce the levels of foulants fed to the membranes.